

ZHUYUAN LIU
PORTFOLIO

Creative Technologist · Designer
MArch · AA School of Architecture

zhuyuan.liu@outlook.com

Physical Computing
Real-Time Systems
Interactive Media
Robotics

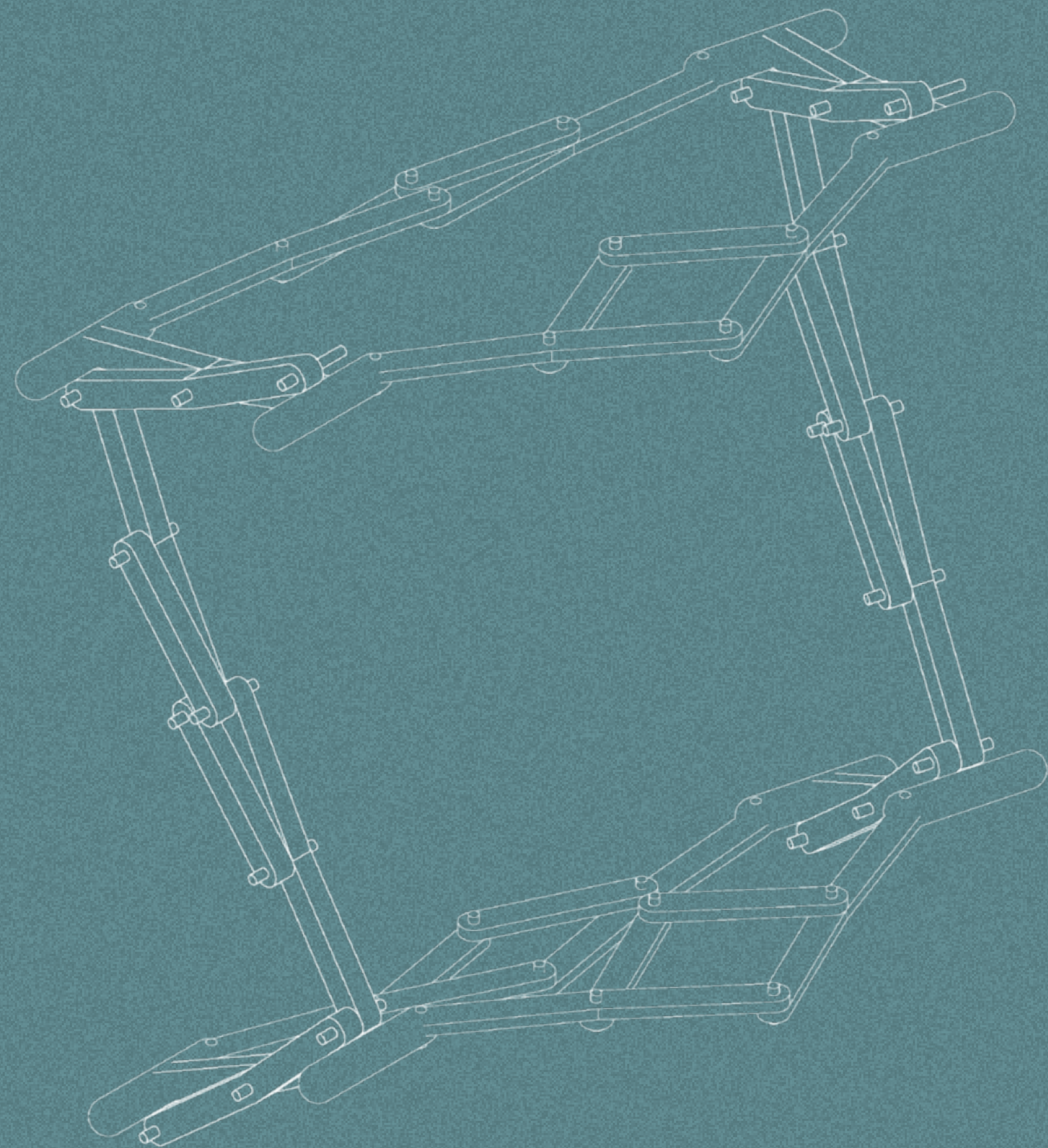
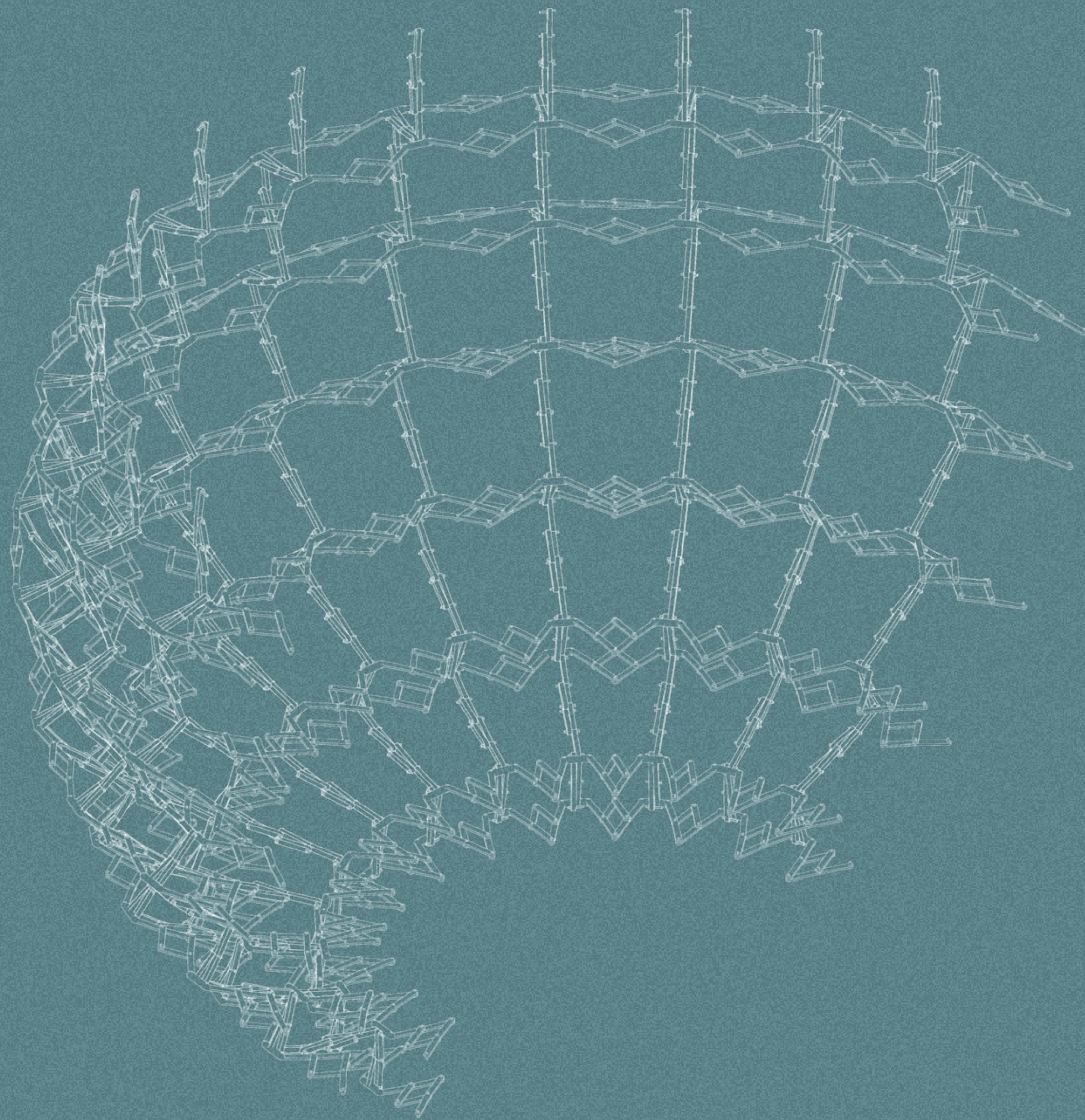


TABLE OF CONTENTS

1. SACRED INTELLIGENCE

INTRODUCTION	03
MODES OF REMOTE PRESENCE	04
ROBOTIC SAND MANDALA	05
AI-DRIVEN RESPONSIVE MURAL	11
PRACTITIONER FEEDBACK	14
SYSTEM OVERVIEW	15

2. LIGHT UP NEW YORK FOR BUT A MOMENT

PROJECT FILM	16
MEDIA BREAKS FREE	17
MEDIA TAKES FORM	18
STRUCTURE, MOVEMENT, FORM	19

3. NO CURTAIN CALL

GAME MAP & SCENES	20
BUILDING THE PERFORMANCE SYSTEM	21

4. MAKING ACROSS SCALES

22

5. PROFESSIONAL PRACTICE

8 CANADA SQUARE	23
1 SPITALFIELDS	24

1. SACRED INTELLIGENCE

REMOTE RITUALS BETWEEN PHYSICAL AND DIGITAL SPACE

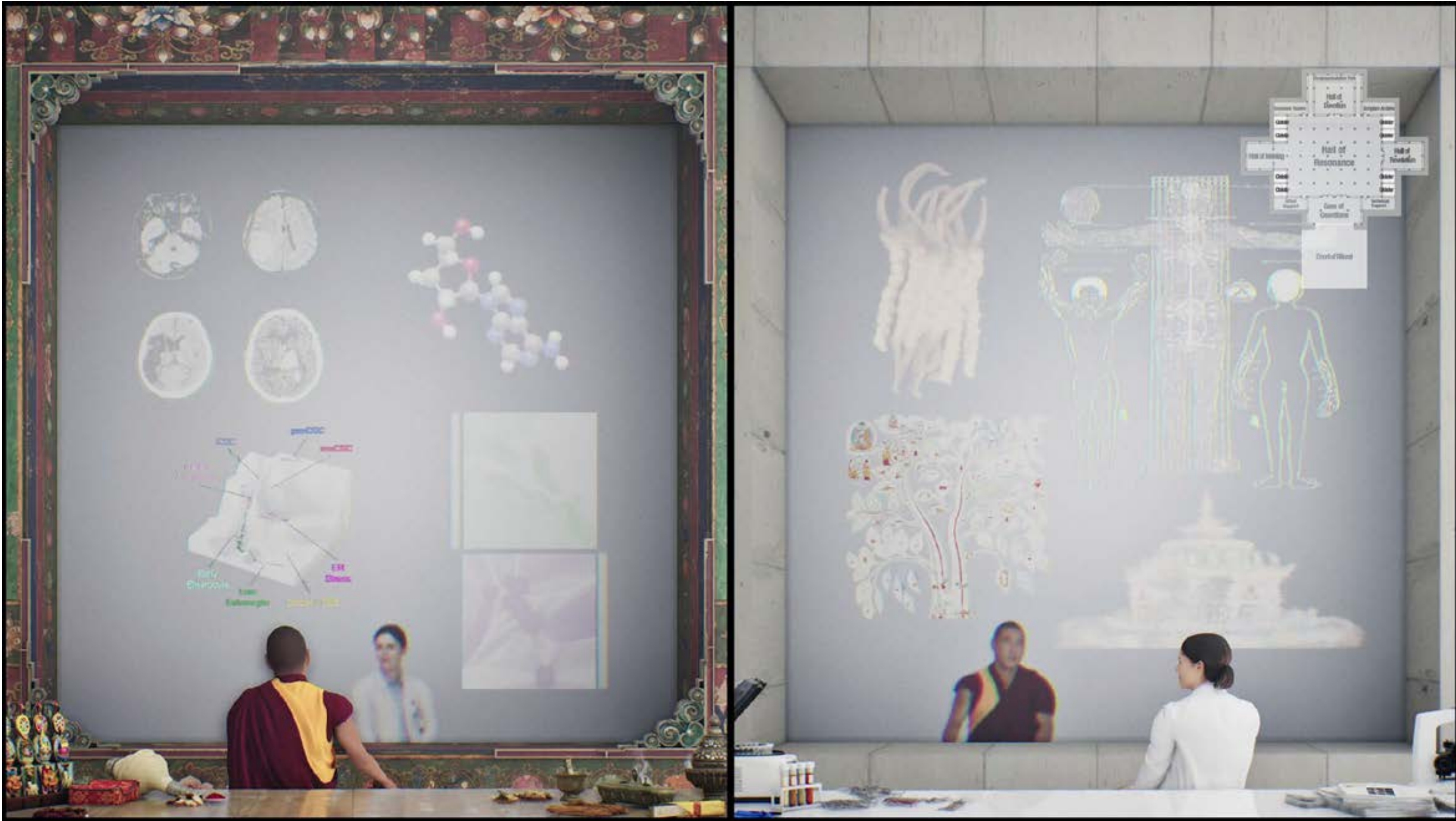
ROBOTICS · AI · PHYSICAL COMPUTING · INTERACTIVE MEDIA

Sacred Intelligence explores how physical ritual and cultural knowledge can be shared across distance through robotics, responsive media and interactive systems. Rather than replacing embodied practice with digital representation, the project builds physical–digital interfaces that allow remote participants to act, make and respond together.

Two core systems were developed and tested as operational 1:1 prototypes.

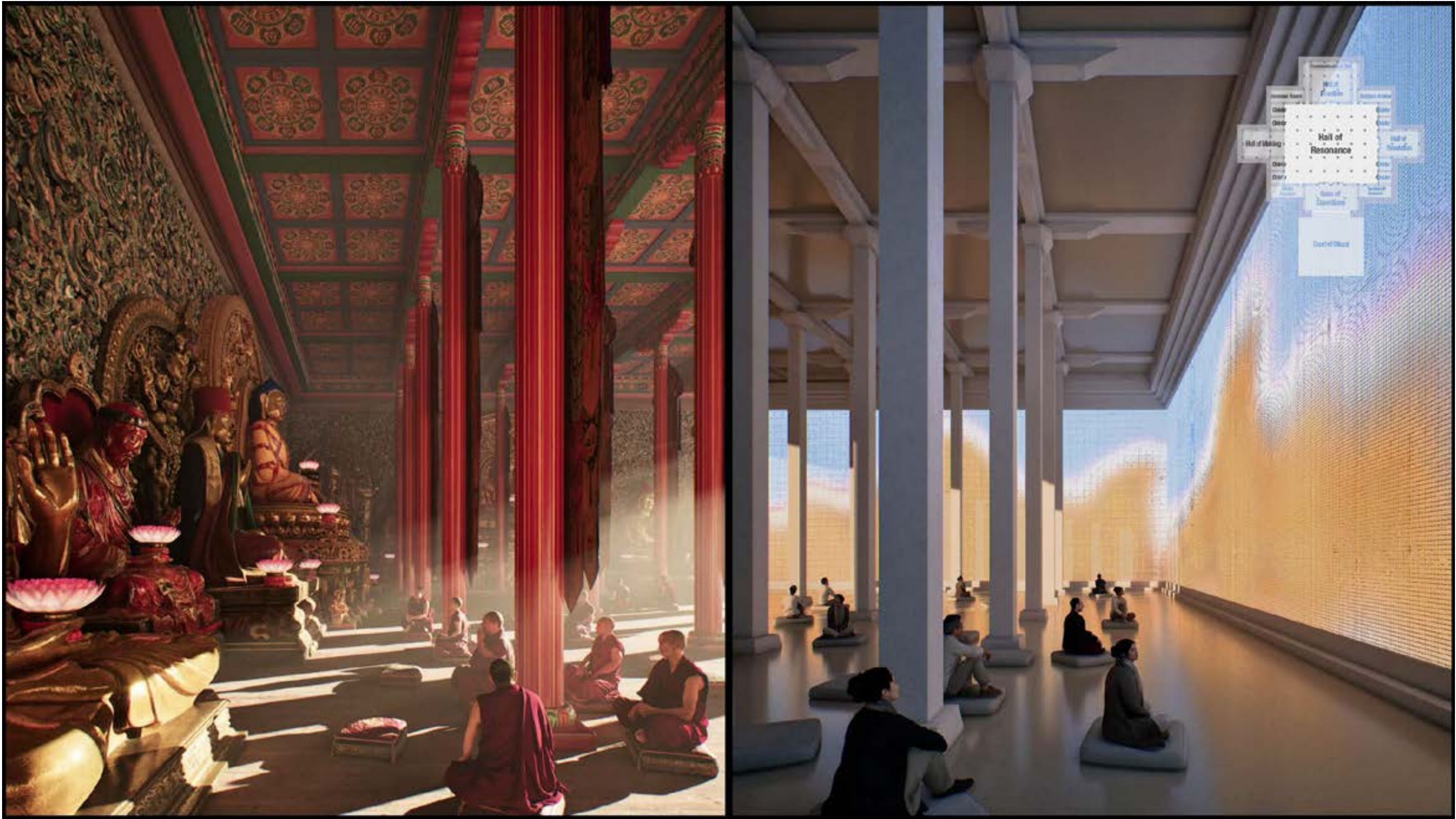


*A Mediated Assembly Hall
Technological Speculation on Tibetan Buddhist Learning Space*



Cloister of Exchange — Shared Knowledge

The Cloister of Exchange connects distant communities through a shared environment for teaching, discussion and study. Rather than treating remote communication as an external screen-based activity, the interaction becomes part of a collective spatial experience.



Hall of Resonance — Shared Presence

The Hall of Resonance explores a quieter form of connection. Breathing, presence and environmental conditions from one location are translated into subtle spatial changes in another, allowing distant participants to sense one another without relying on direct visual communication.

ROBOTIC SAND MANDALA

FINAL EXPERIENCE + PHYSICAL PROOF



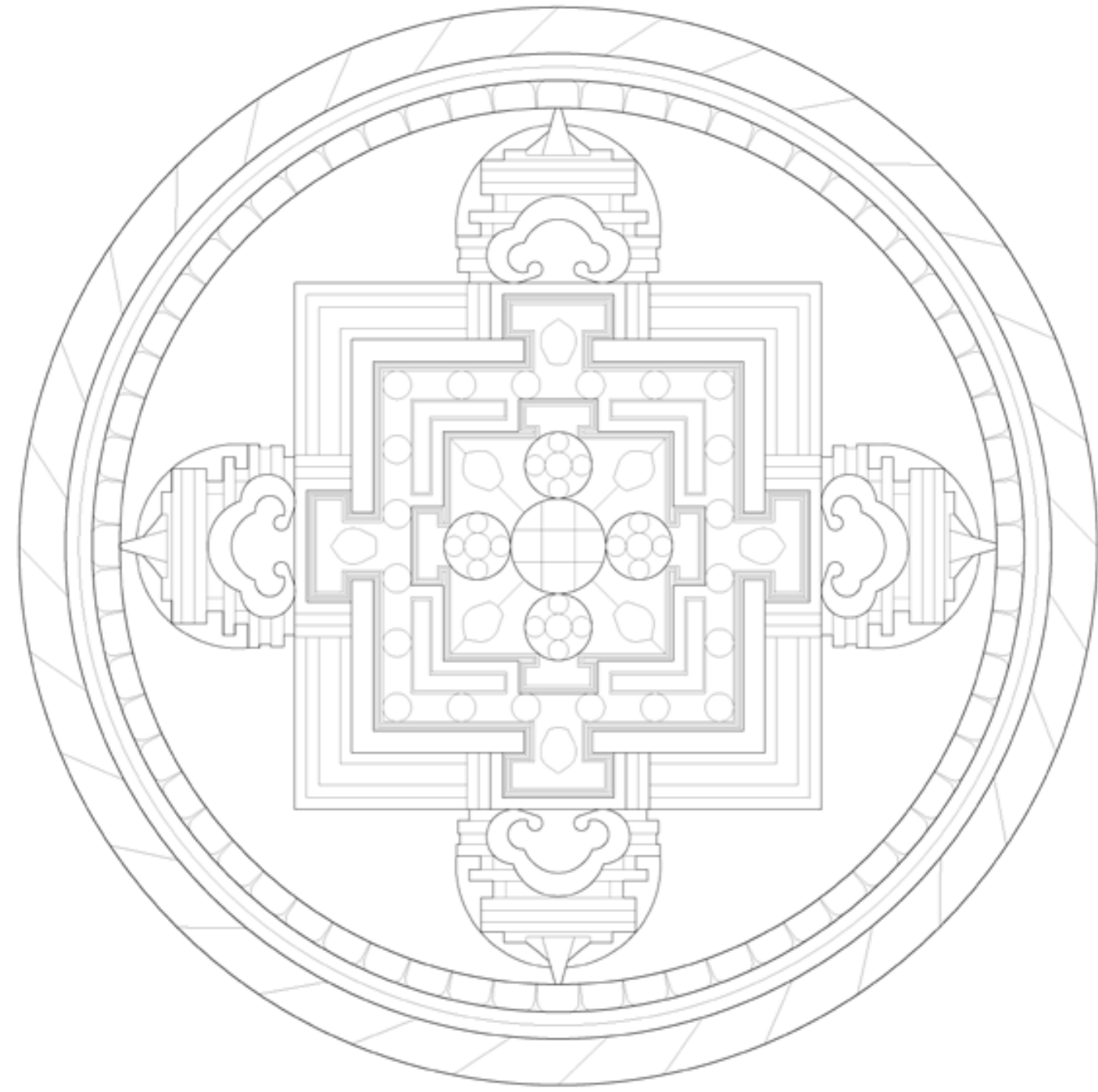
Remote Ritual as Physical Making

During the annual sand mandala ritual, drawing instructions can be transmitted from a mandala master at Gonggar to a robotic installation in London. Rather than observing a digital reproduction, remote participants engage with the same process of making, extending an embodied practice across geographical distance.



Multi-Colour Sand Deposition Prototype

The 1:1 prototype integrates four independent sand channels with controlled colour switching, regulated outlets and a soft-brush tip. Through repeated physical testing, the system developed into a stable method for multi-colour robotic sand deposition rather than remaining a digital simulation.



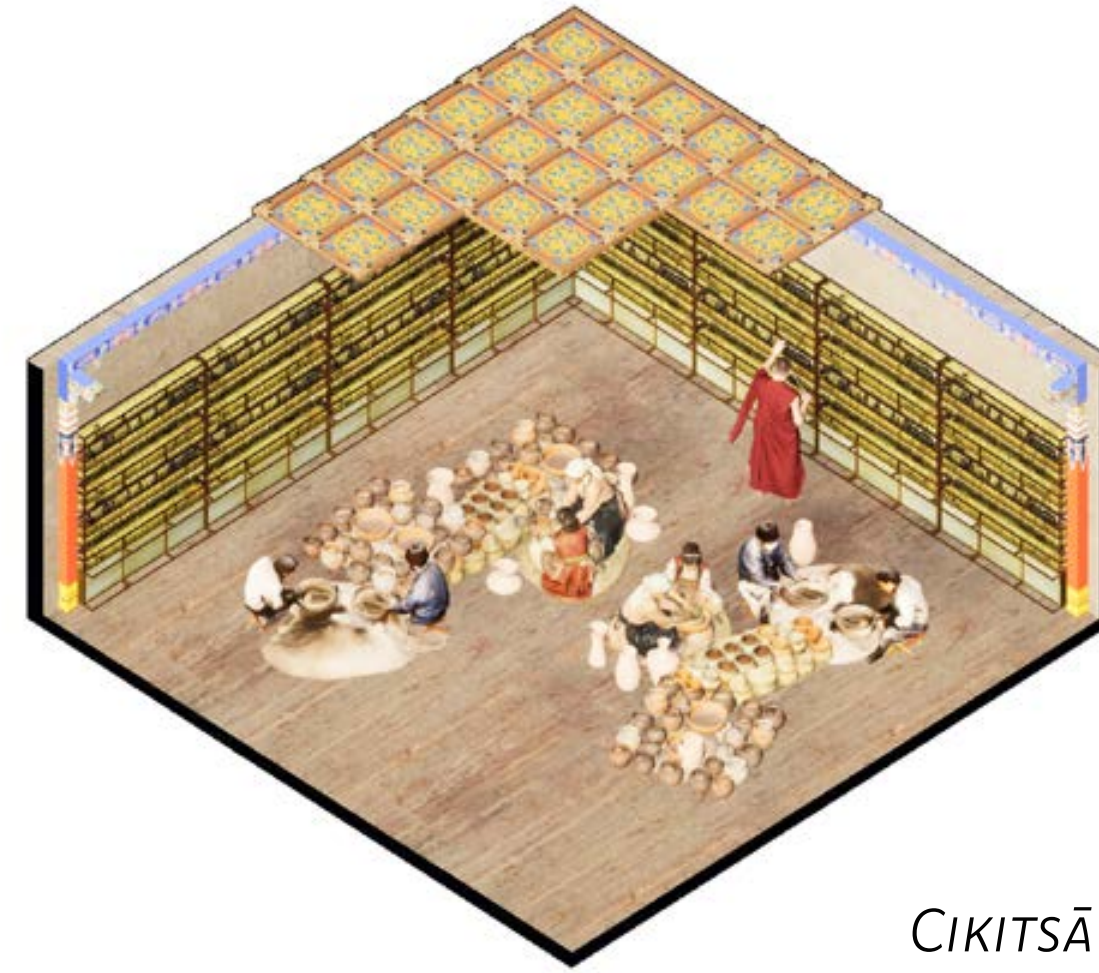
Mapping a Modular Knowledge System

The project begins with the Five Sciences, a traditional framework through which Tibetan Buddhist knowledge and practice are organised. Medicine, language, arts and crafts, inner science and logic are treated as interconnected domains rather than isolated subjects.

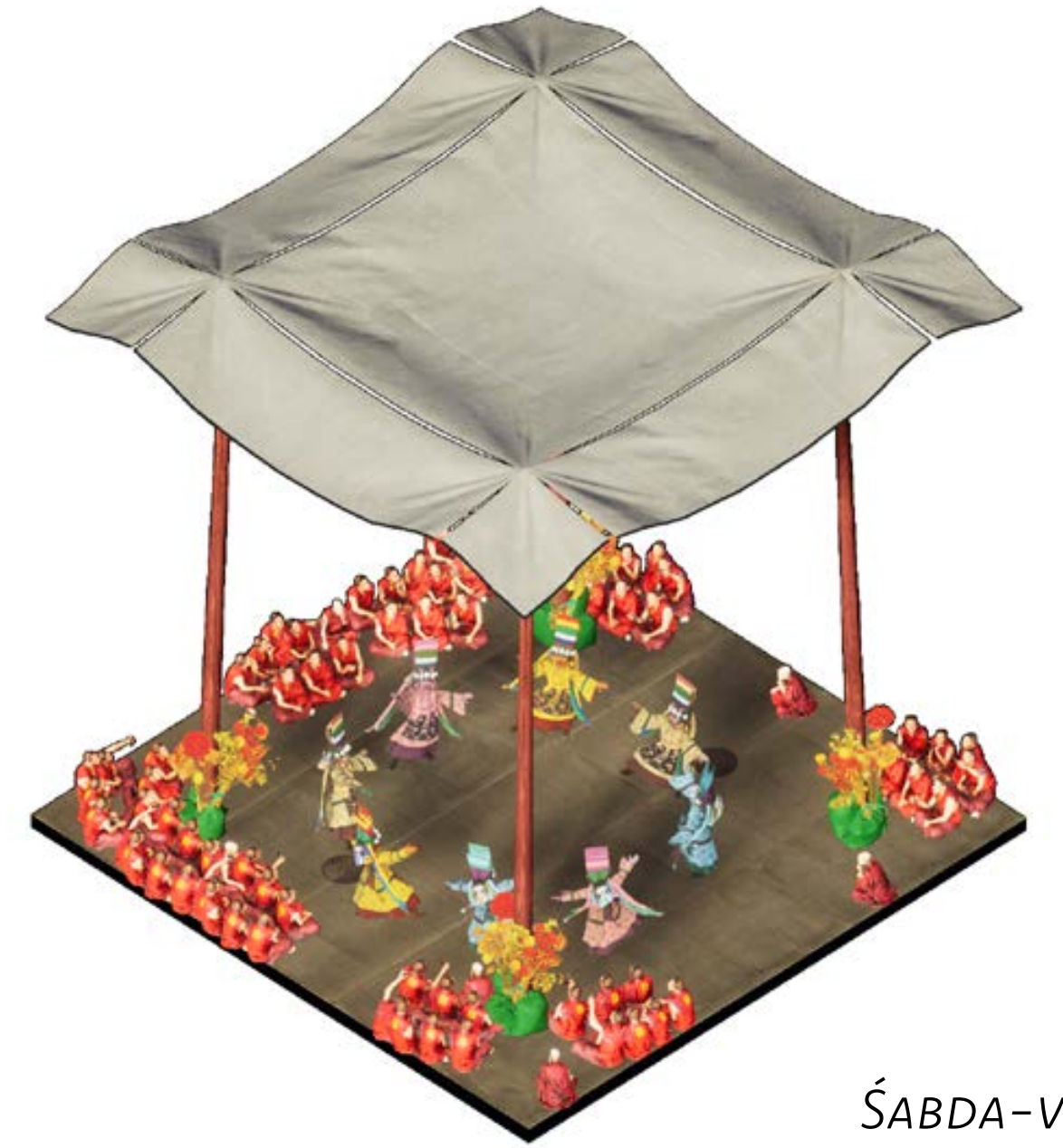
From this wider system, the sand mandala is selected for deeper study as a practice whose geometry, construction sequence and symbolic organisation can be progressively analysed, decomposed and translated into computational form.



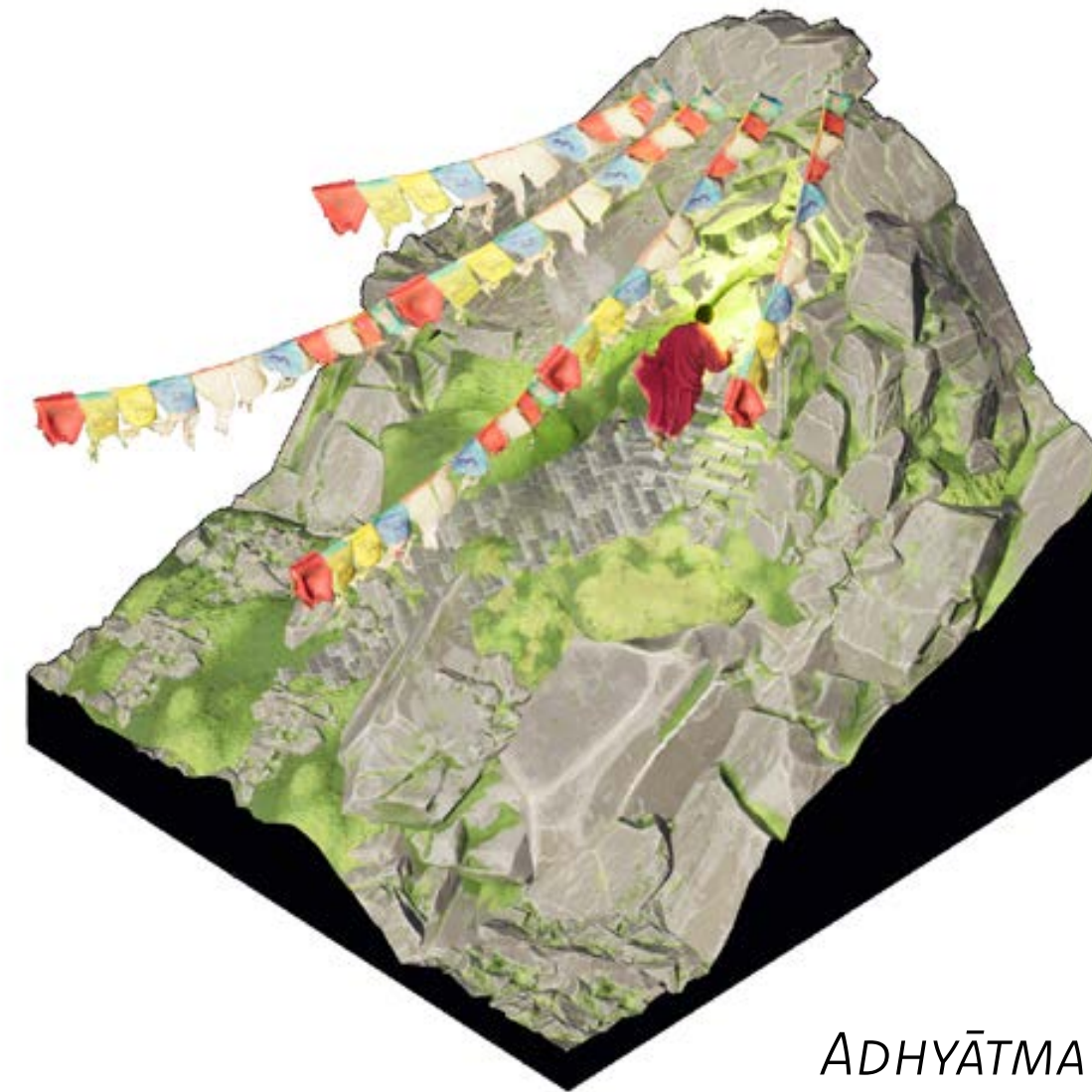
ŚILPA-VIDYĀ
"SCIENCE OF ARTS AND CRAFTS"



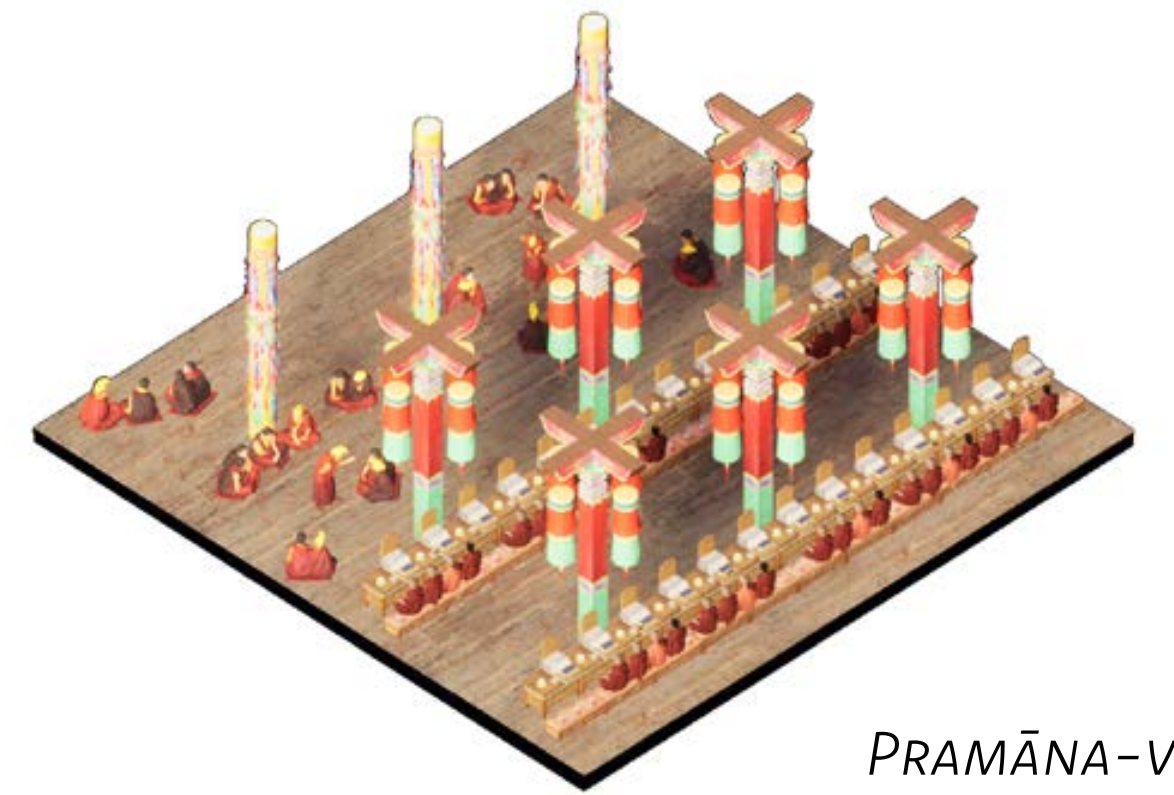
CIKITSĀ-VIDYĀ
"SCIENCE OF MEDICINE"



ŚABDA-VIDYĀ
"SCIENCE OF LANGUAGE"



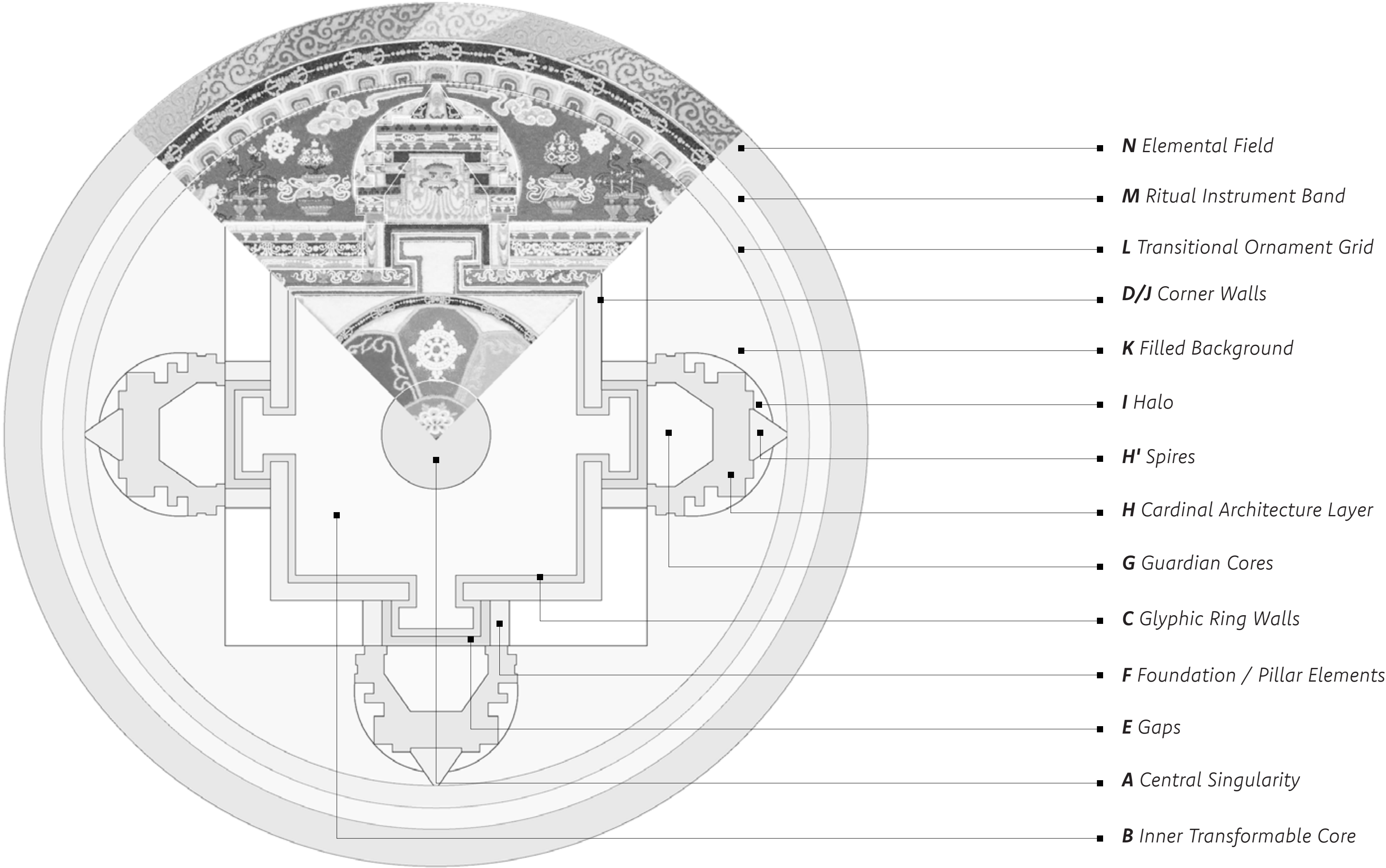
ADHYĀTMA-VIDYĀ
"INNER SCIENCE"



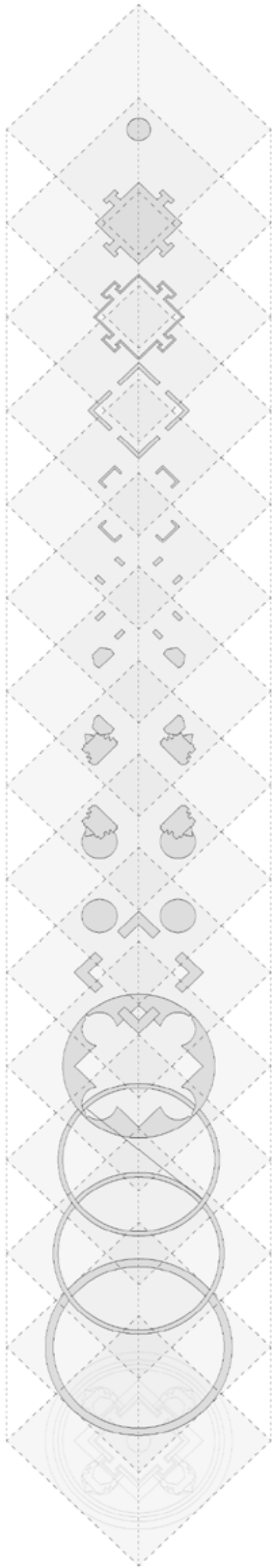
PRAMĀNA-VIDYĀ
"SCIENCE OF LOGIC"

From Mandala to Data – Component Decomposition & Geometric Encoding

Rather than treating the mandala as a single image, recurring centres, boundaries, gateways, walls and layered enclosures are separated into a consistent component system. Each element is assigned a position within the wider hierarchy before being translated into structured geometric data. Component IDs, bounding regions, centres, polygon counts and internal voids allow the visual composition to become editable and machine-readable while retaining the relationships between its parts.



COMPONENT ID · AREA · GEOMETRIC CENTRE · POLYGON COUNT · INTERNAL VOIDS



A
B
C
D
E
F
G
H
I
J
K
L
M
N

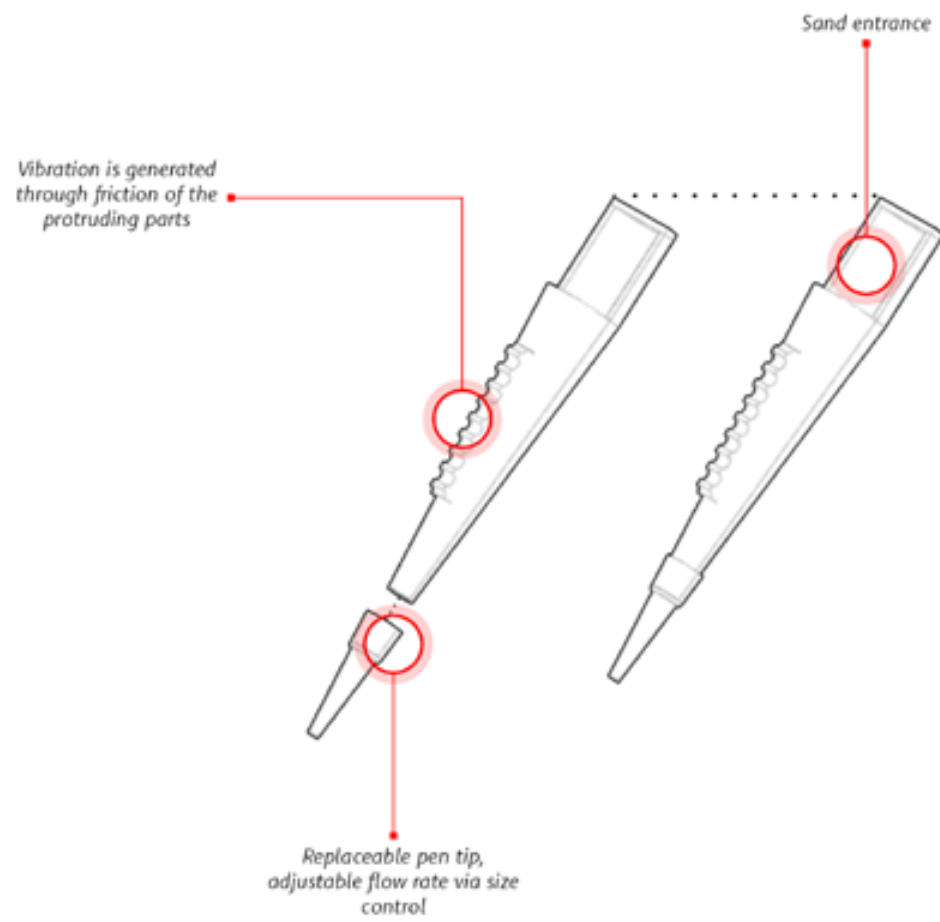


Build • Test • Refine — I

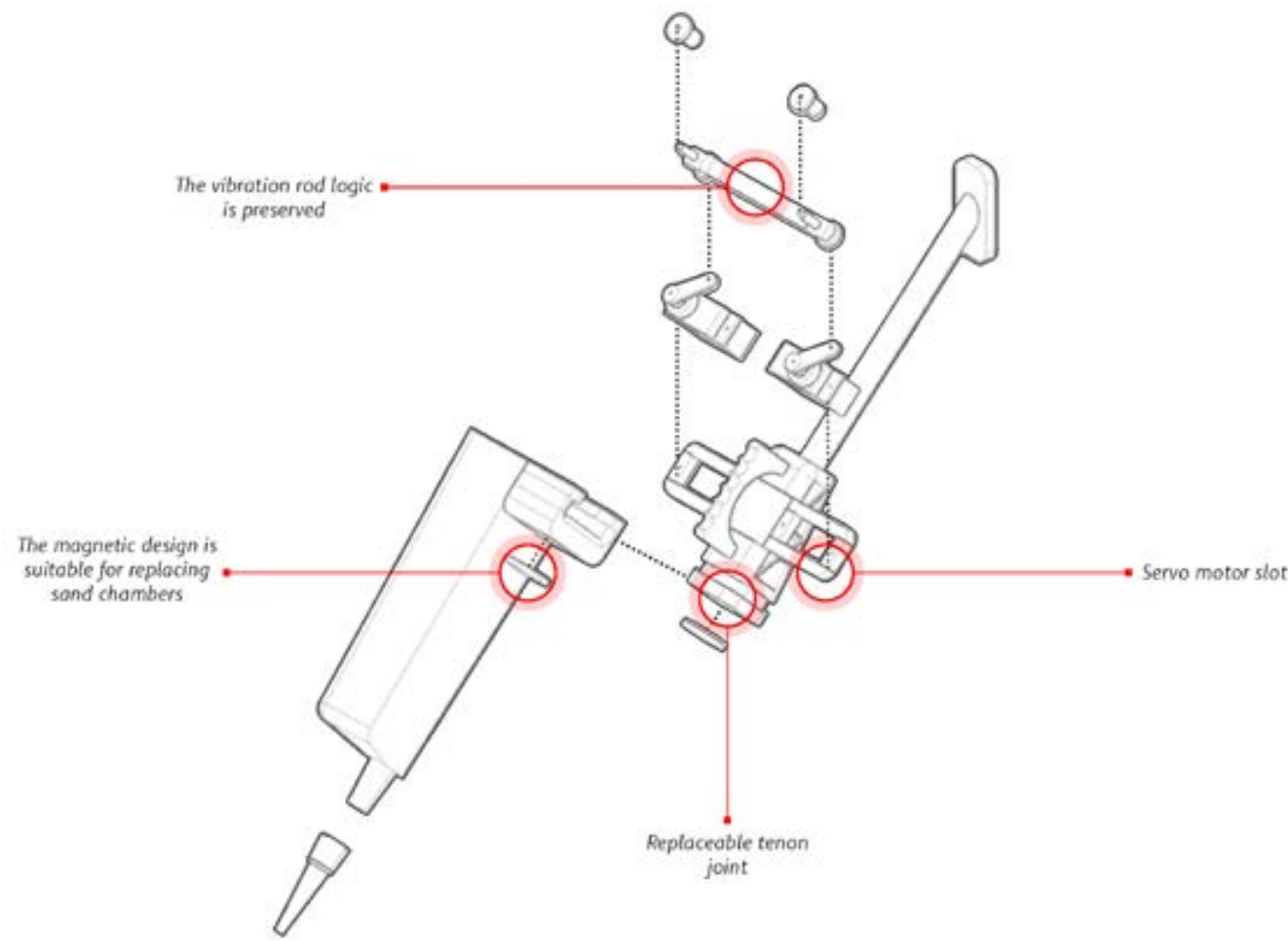
The first prototypes began by reinterpreting the traditional friction-vibration principle of sand mandala drawing for robotic use. Early iterations tested how friction, vibration and tool geometry could generate a controllable sand flow, progressing from a modernised hand tool to a lever-based end-effector designed for robotic mounting.

These experiments established the basic relationship between material behaviour, dispensing mechanics and robotic movement.

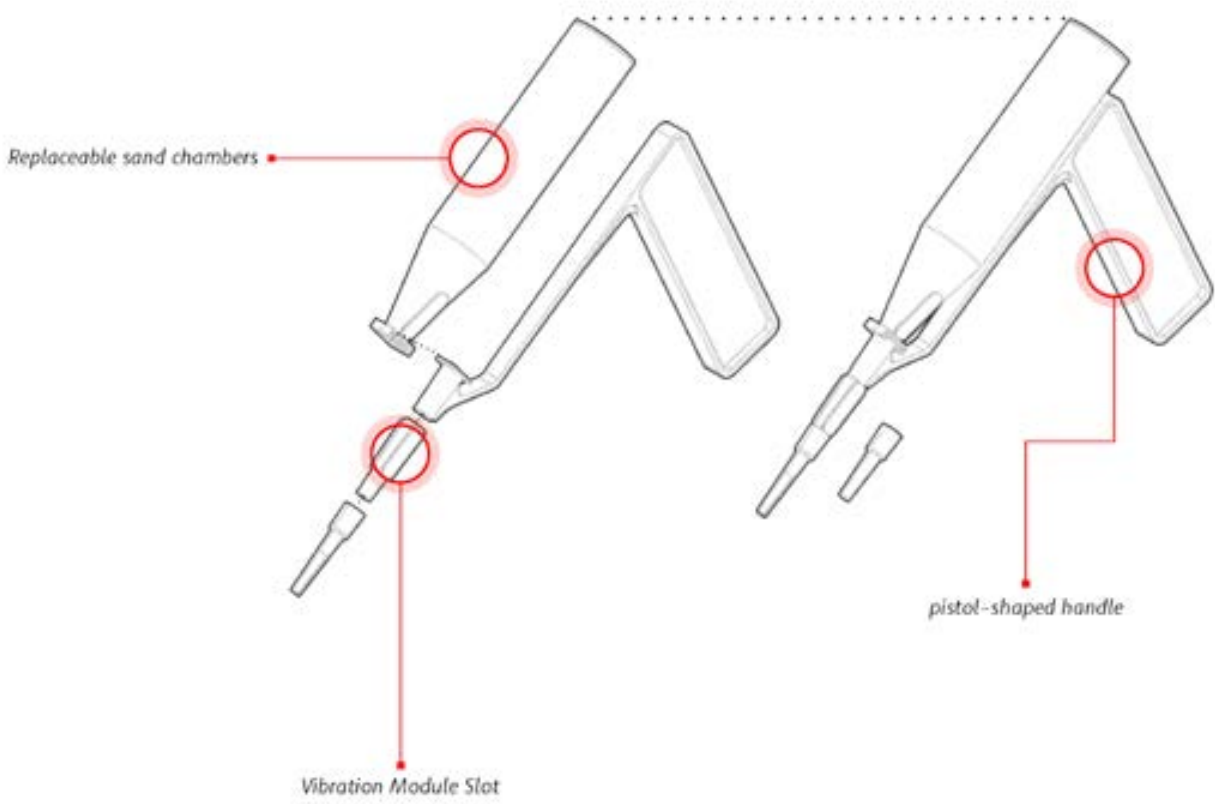
PROTOTYPE 01 — TRADITIONAL FRICTION TOOL



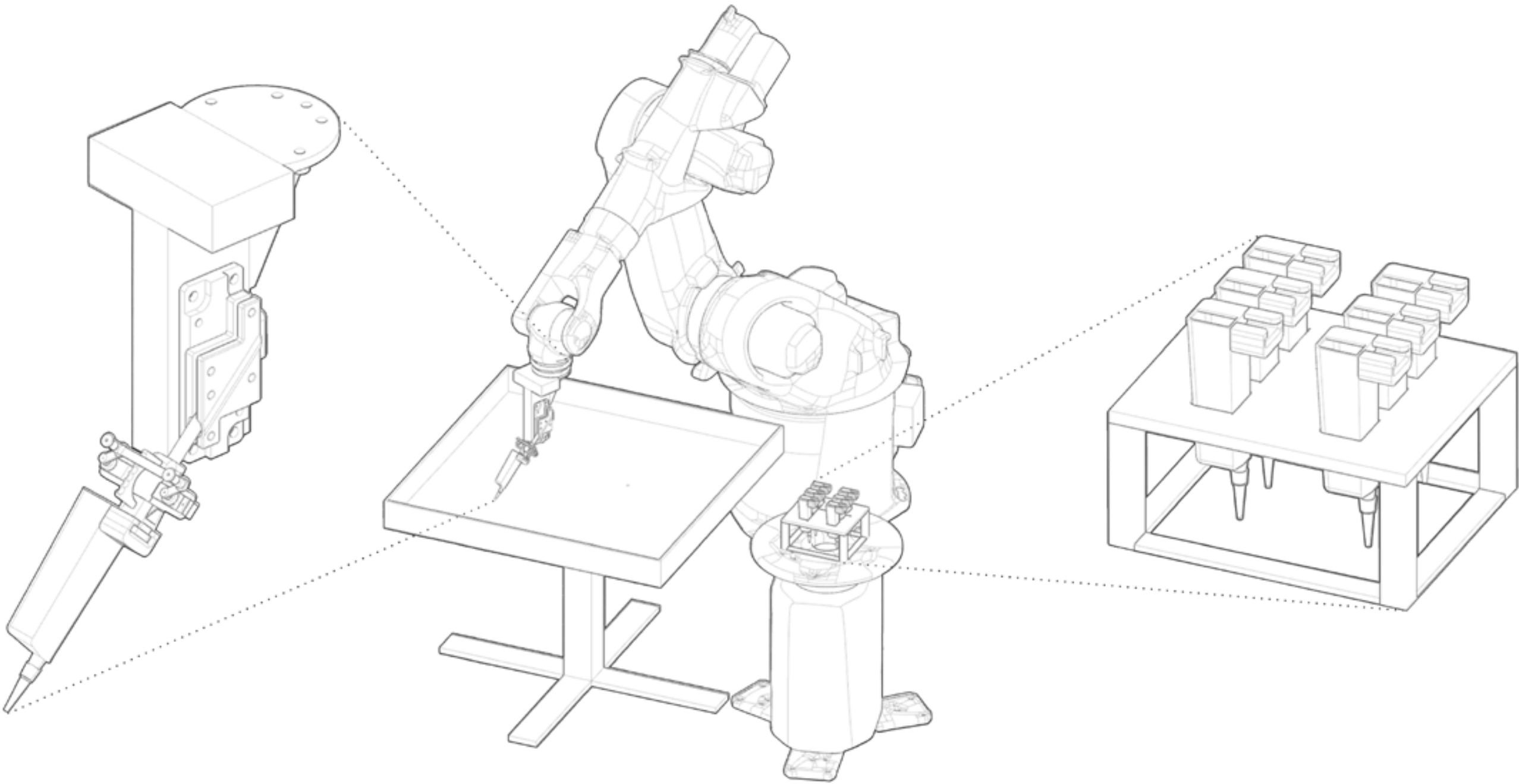
PROTOTYPE 02 — LEVER FRICTION END-EFFECTOR



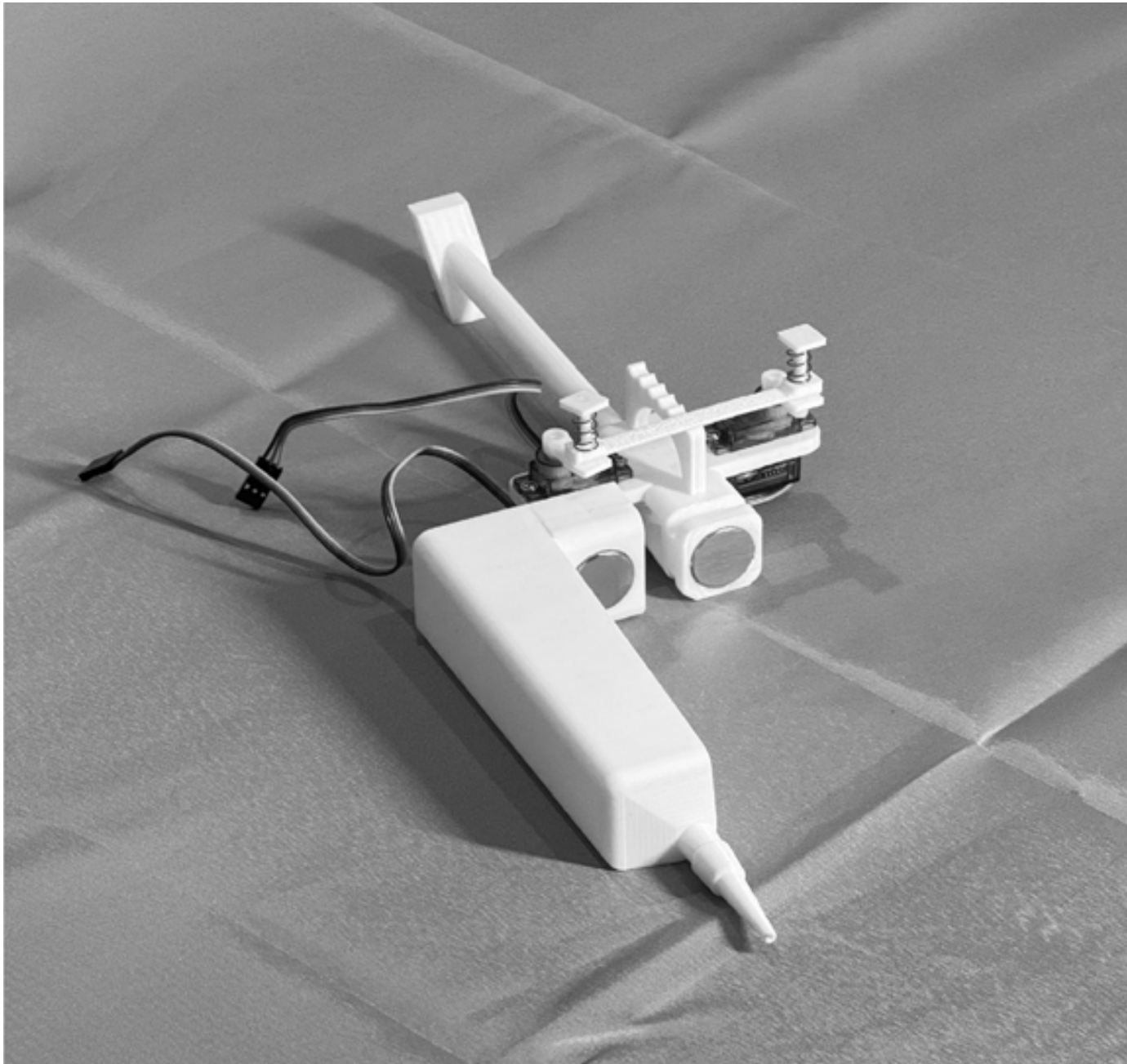
PROTOTYPE 03 — HANDHELD VIBRATORY SAND GUN



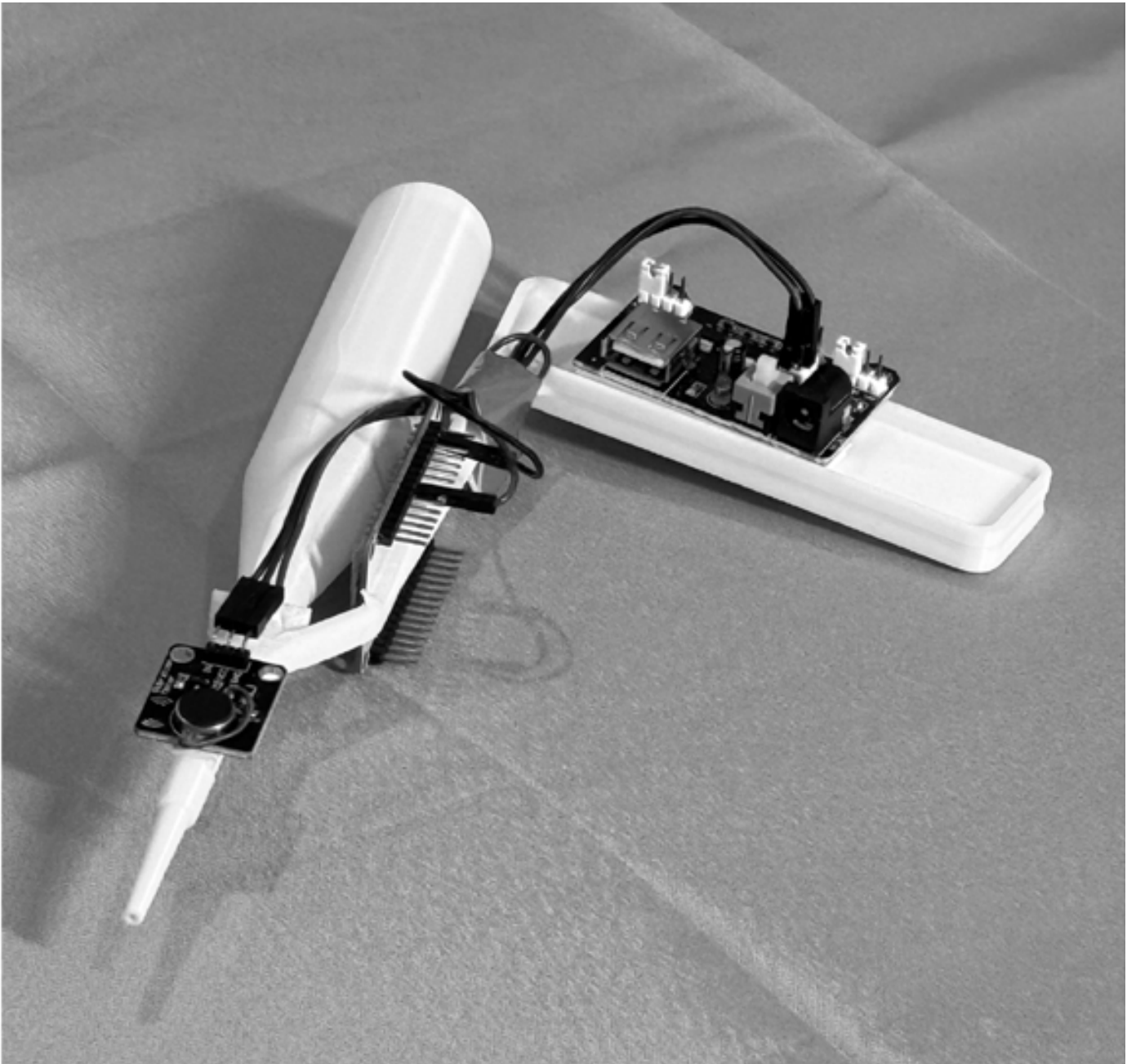
ROBOTIC MOUNTING TESTS



PROTOTYPE 02 - PHYSICAL MODEL



PROTOTYPE 03 - PHYSICAL MODEL

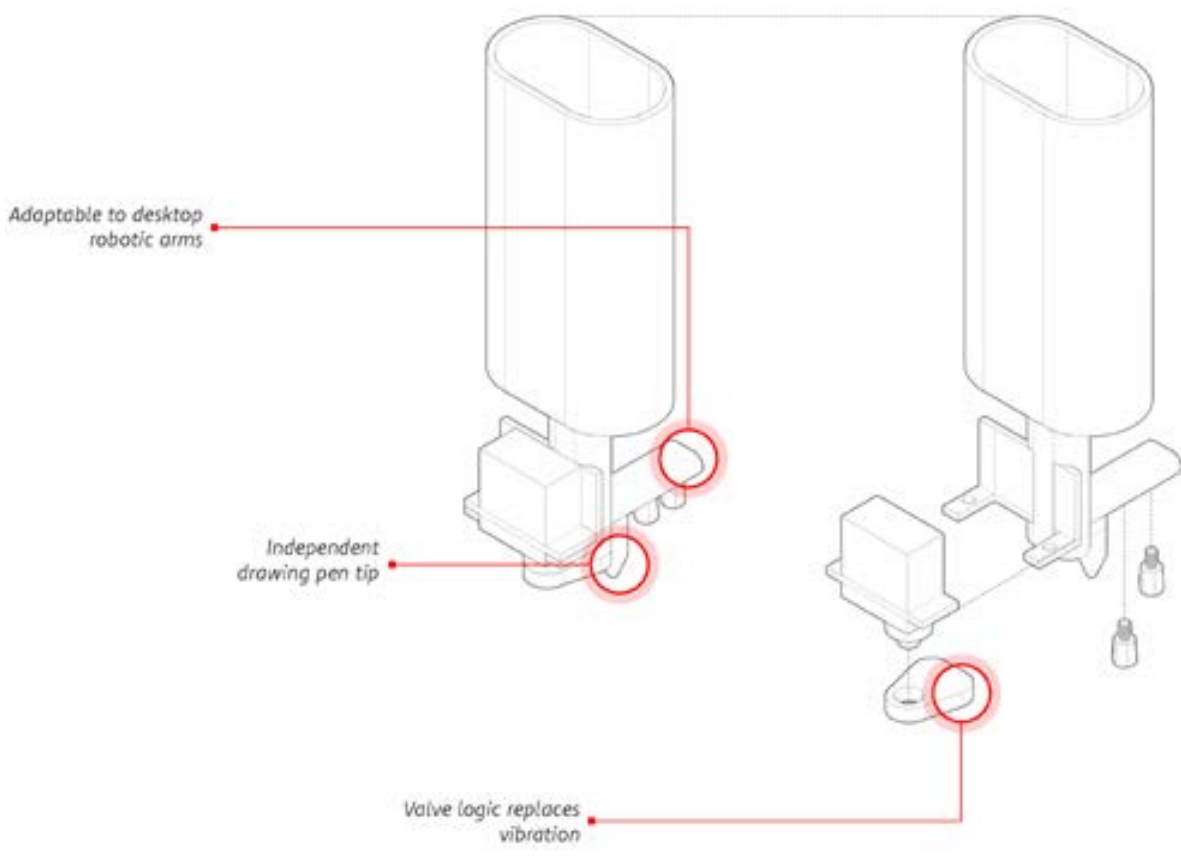


Build • Test • Refine — II

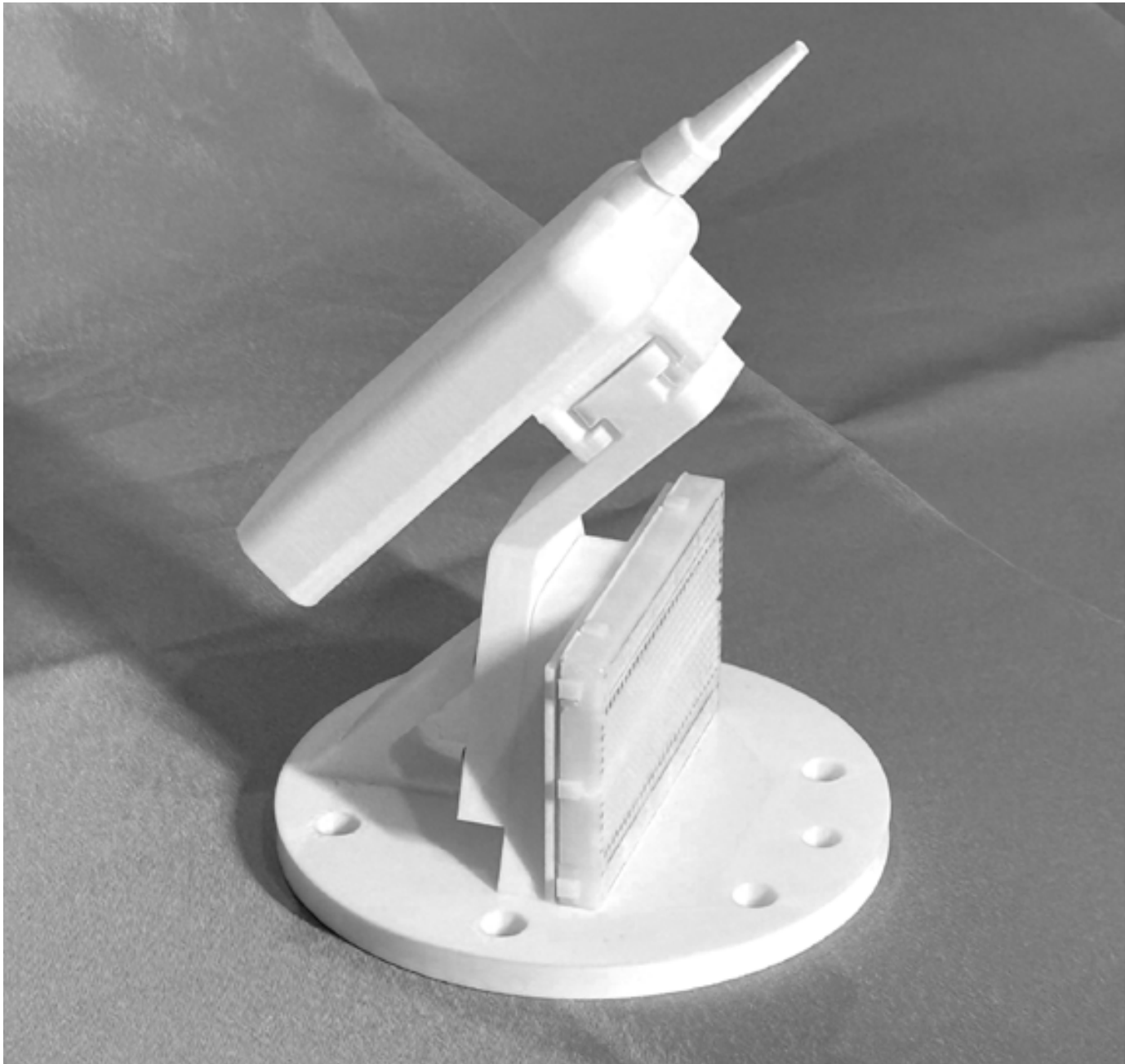
Later iterations focused on achieving more stable and controllable deposition. Angle-based flow control was first tested through a high-capacity dispenser, followed by two generations of valve-controlled heads combining regulated sand release with a drawing stylus.

The final iteration integrated multiple switchable sand channels into a single robotic end-effector, enabling controlled multi-colour deposition during operation.

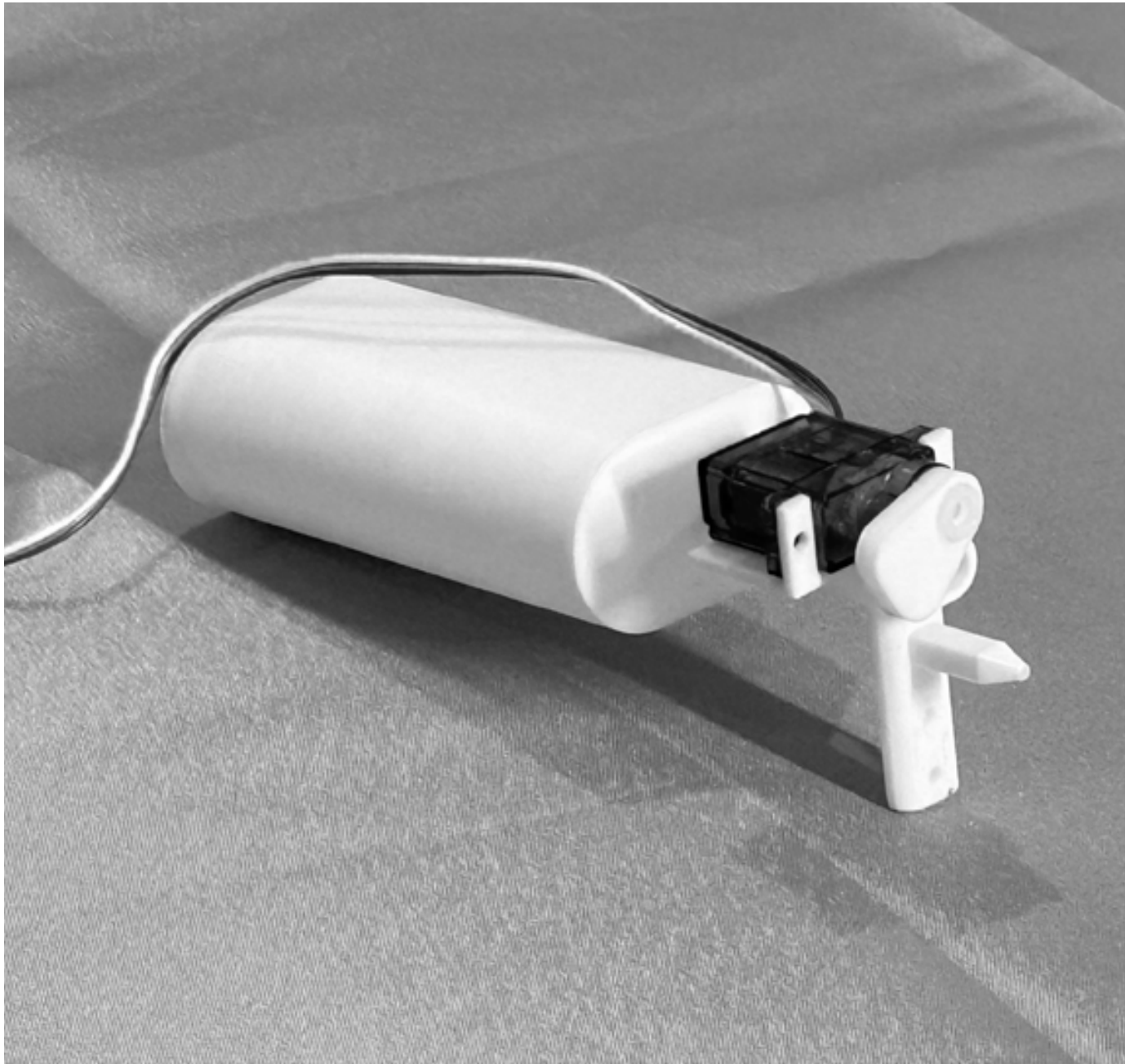
PROTOTYPE 05 — VALVE-CONTROLLED DEPOSITION HEAD, GEN 1



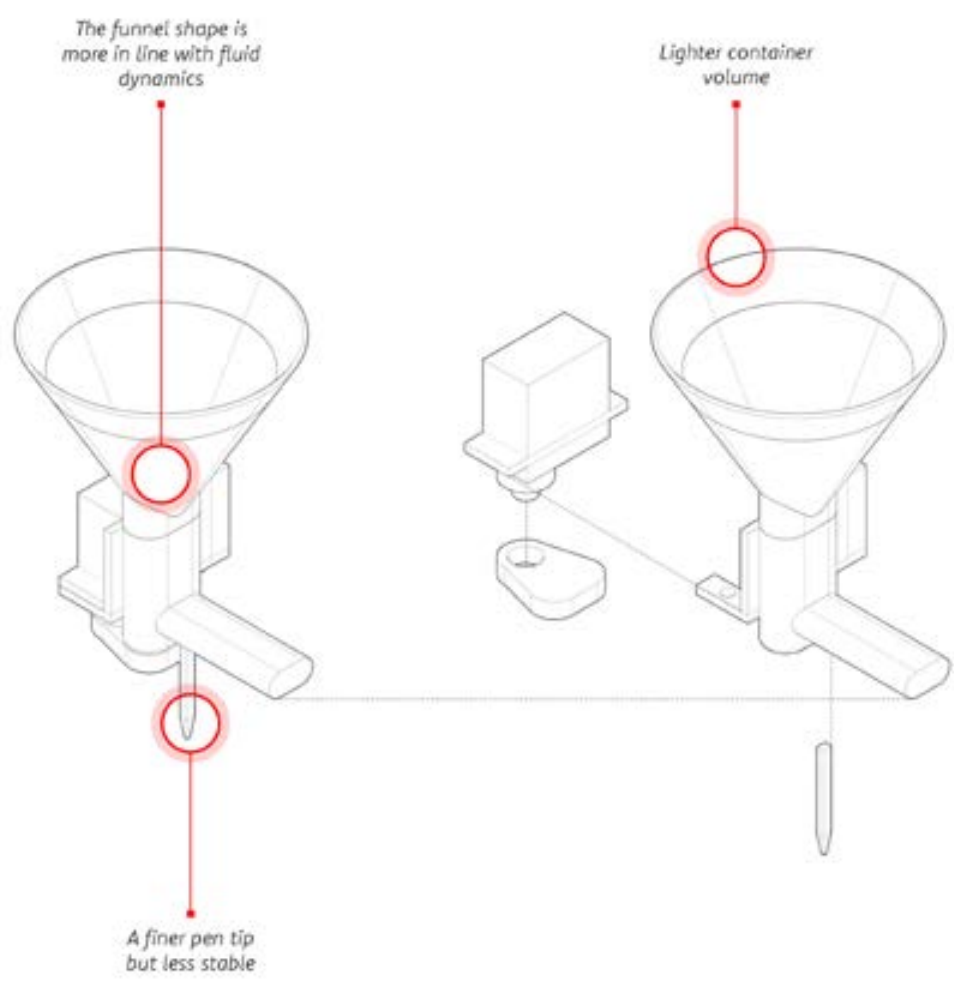
PROTOTYPE 04 — ANGLE-CONTROLLED HIGH-CAPACITY DISPENSER



PROTOTYPE 05 - PHYSICAL MODEL



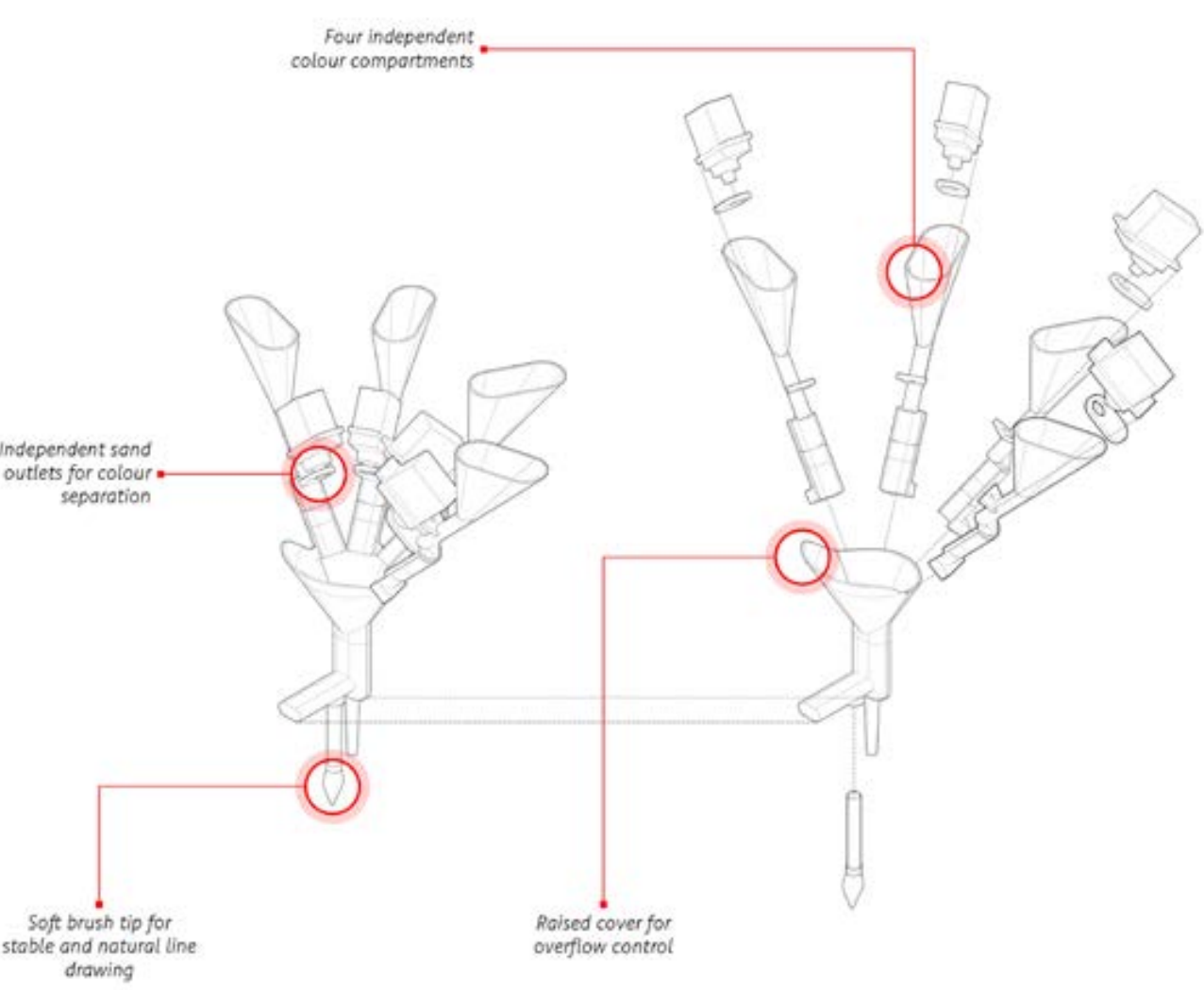
PROTOTYPE 06 — VALVE-CONTROLLED DEPOSITION HEAD, GEN 2



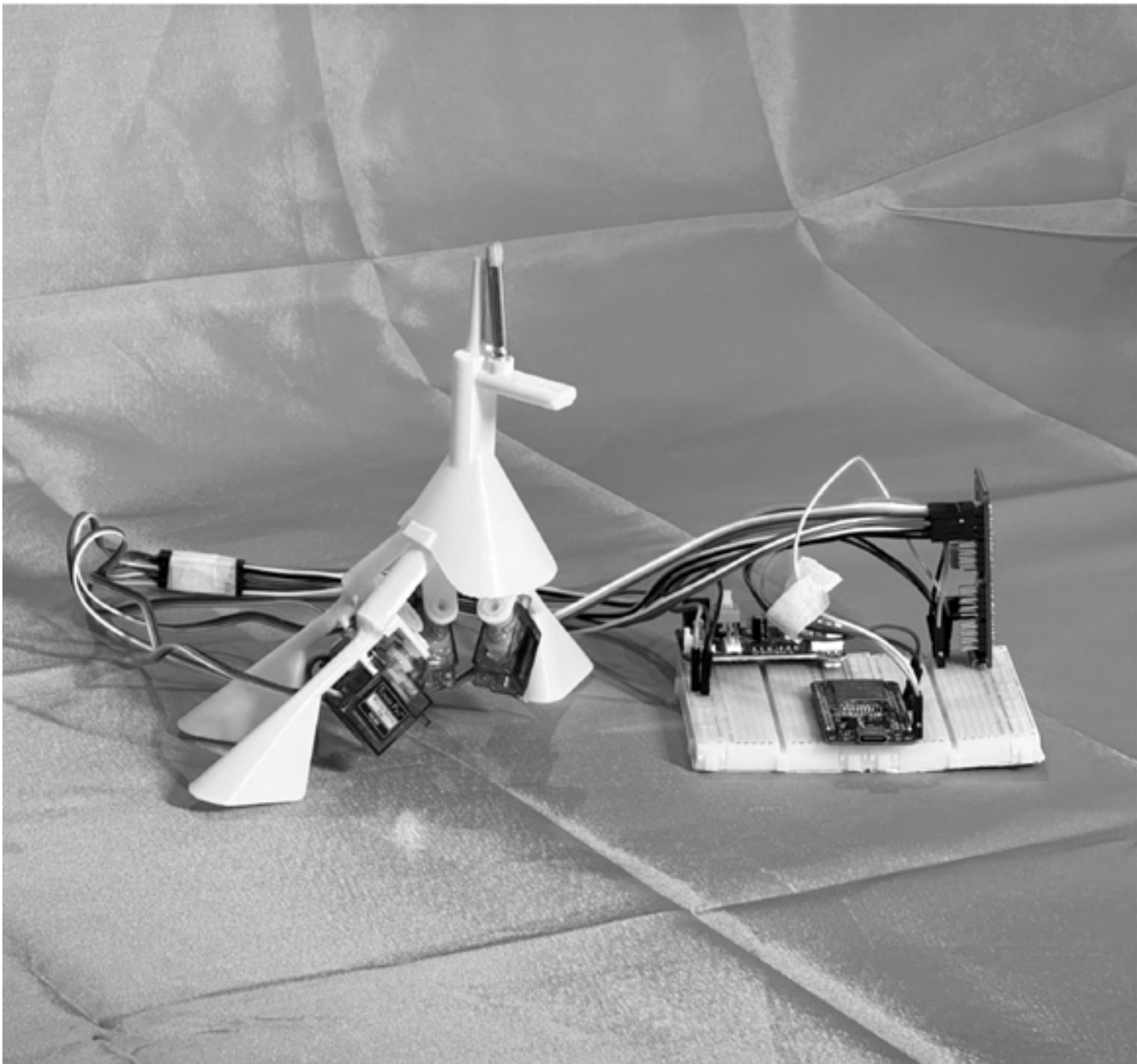
PROTOTYPE 06 - PHYSICAL MODEL



PROTOTYPE 07 — MULTI-COLOUR ROBOTIC SAND DEPOSITION HEAD



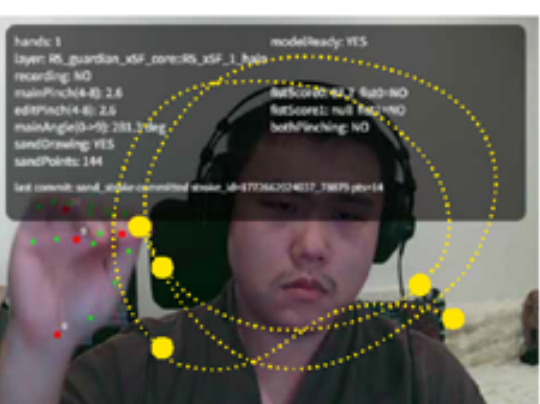
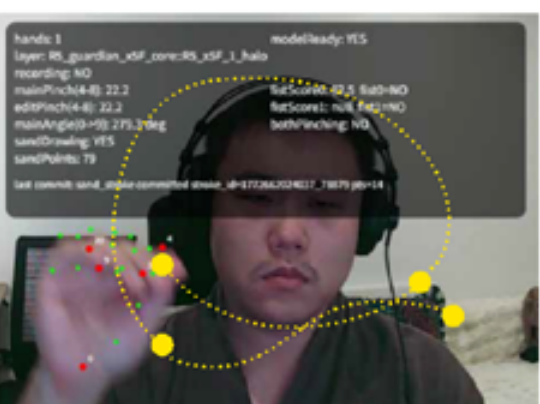
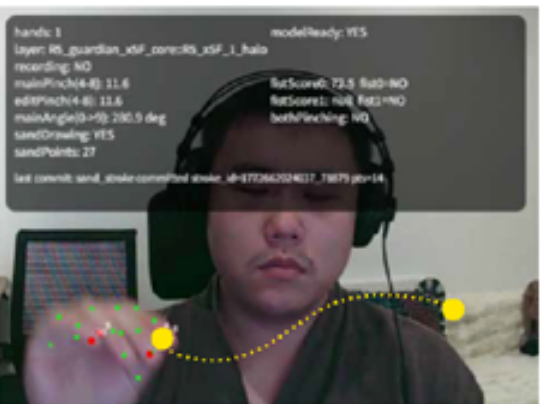
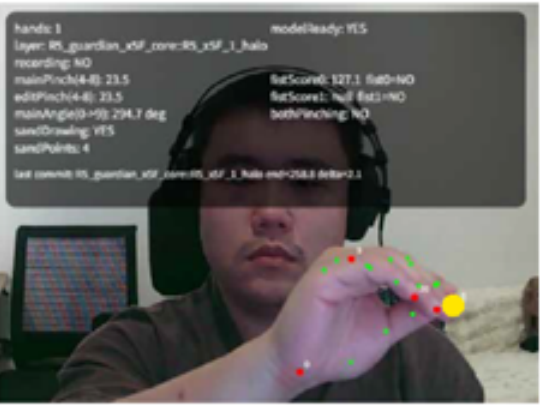
PROTOTYPE 07 - PHYSICAL MODEL



Making the System Playable

Gesture-Based Human–Machine Interaction

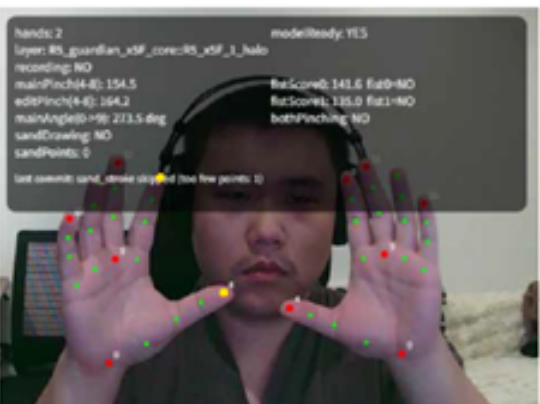
Single-hand pinch and drag → sand deposition, controlling the placement and movement of sand in real time.



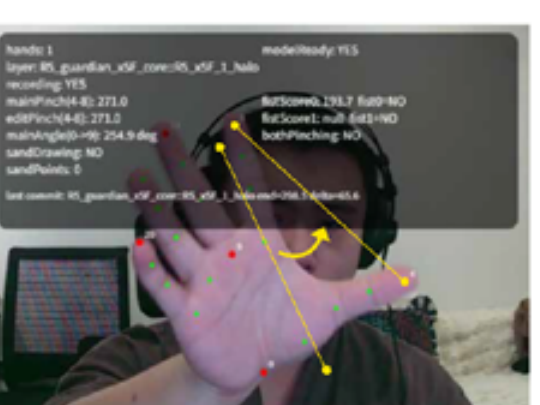
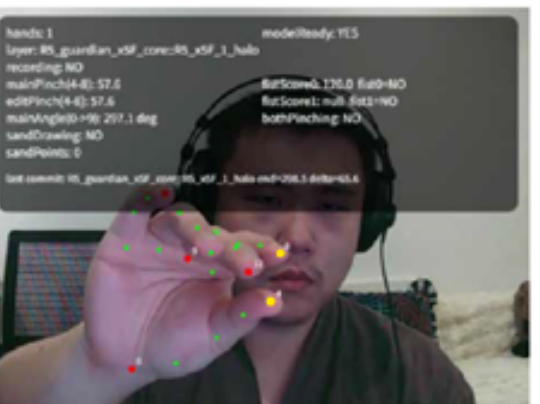
Both hands clenched (fists) → execute Draw All, triggering the full drawing sequence.



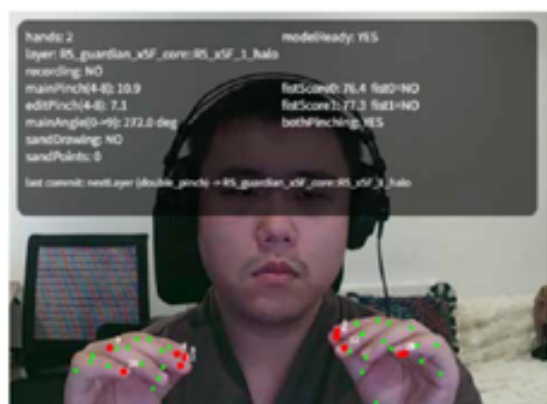
Double-hand rotation around an anchor point → edit stroke paths, enabling adjustments to existing trajectories.



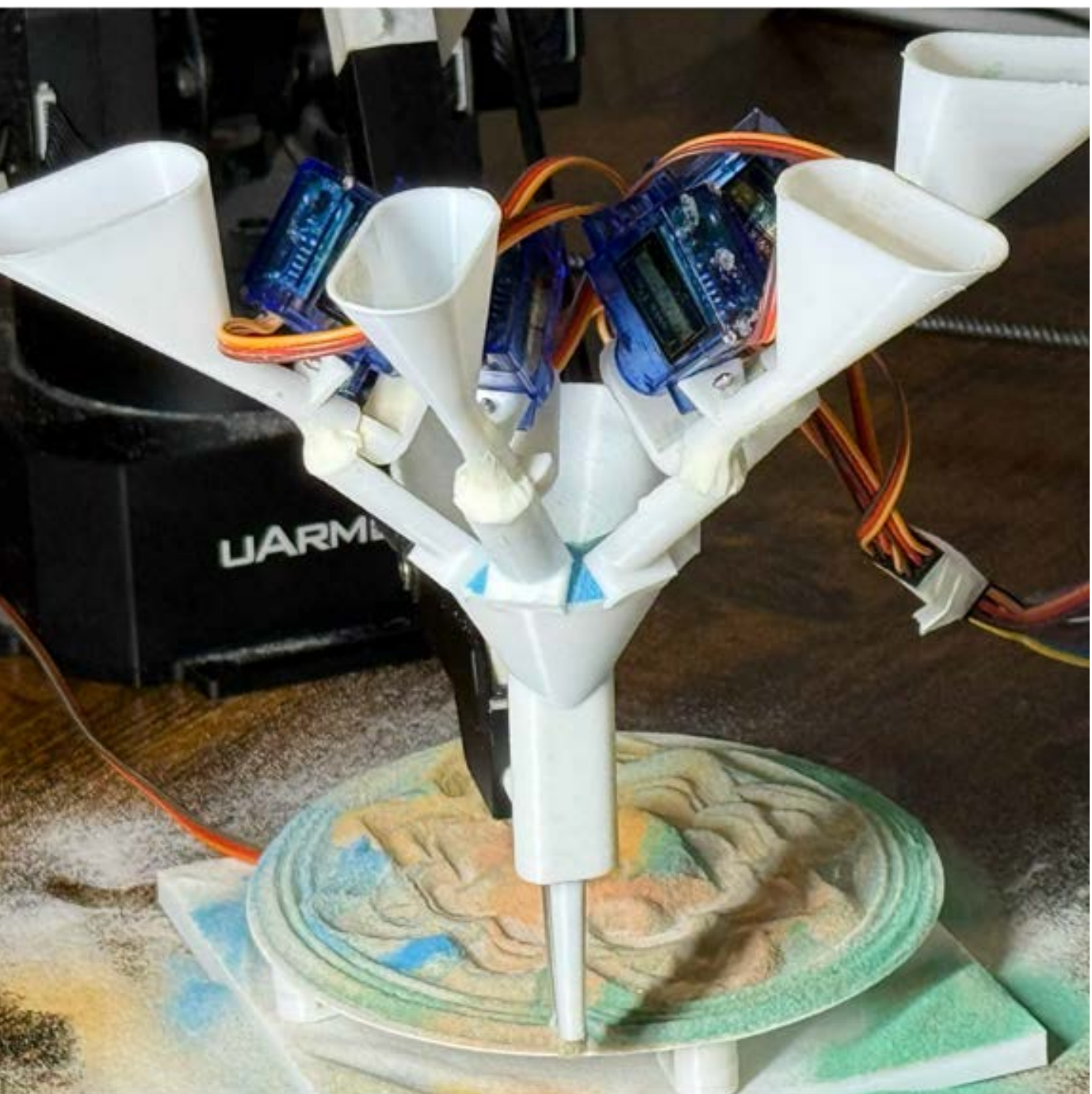
Single-hand rotation around an anchor point → edit stroke paths, enabling adjustments to existing trajectories.



Both hands pinching → switch layers, allowing the user to move between different drawing layers.



Command Timestamp		Active Layer	Rotation Angle	Rotation Delta	Execution Command	Instruction Payload	System Trace	Stroke Points	Execution Status		
1	ts	A	B	C	D	E	F	G	H	I	J
1	2020-02-11T19:44:19.519Z	layer		angle_deg	delta_deg	action	payload	log	points_n	bytes	processed
2	2020-02-11T19:44:21.552Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		24	1918	1
3	2020-02-11T19:44:21.552Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		8	723	1
4	2020-02-11T19:44:23.576Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		30	2405	1
5	2020-02-11T19:44:24.903Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		3	359	1
6	2020-02-11T19:44:26.106Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		10	854	1
7	2020-02-11T19:44:28.382Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		21	1705	1
8	2020-02-11T19:44:30.216Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		28	2231	1
9	2020-02-11T19:45:38.294Z	R1_ring		237.7889902	-48.62331026	log	["layer":"R1_ring","angle_d:start=286.4lend=237.8delta=-48.6"]			139	1
10	2020-02-11T19:45:38.032Z	R1_ring		325.1842875	79.7661969	log	["layer":"R1_ring","angle_d:start=245.4lend=325.2delta=79.8"]			135	1
11	2020-02-11T19:45:40.380Z	ALL		0	0	draw_all	["layer":"__ALL__","angle_i:gesture=double_fist"]			95	1
12	2020-02-11T20:28:31.954Z	R1_ring		257.1400985	9.807989175	log	["layer":"R1_ring","angle_d:start=247.3lend=257.1delta=9.8"]			134	1
13	2020-02-11T20:28:33.385Z	R1_ring		259.7128390	1.056257742	log	["layer":"R1_ring","angle_d:start=258.7lend=259.7delta=1.1"]			134	1
14	2020-02-11T20:28:39.176Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		92	7070	1
15	2020-02-11T20:45:41.254Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		89	6806	1
16	2020-02-11T20:45:44.484Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		19	1681	1
17	2020-02-11T21:26:08.216Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		79	8128	1
18	2020-02-11T21:27:02.499Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		50	4019	1
19	2020-02-11T21:27:32.670Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		19	1650	1
20	2020-02-11T21:27:57.381Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		42	3347	1
21	2020-02-11T21:28:18.773Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		44	3550	1
22	2020-02-12T03:12:46.715Z	R1_ring		240.2319011	-36.70206029	log	["layer":"R1_ring","angle_d:start=278.0lend=240.2delta=-36.7"]			138	1
23	2020-02-12T03:15:28.095Z	R1_ring		242.5879617	-30.70349631	log	["layer":"R1_ring","angle_d:start=273.3lend=242.6delta=-30.7"]			138	1
24	2020-02-12T03:33:51.357Z	R1_ring		227.6446198	-47.66396629	log	["layer":"R1_ring","angle_d:start=275.3lend=227.6delta=-47.7"]			138	1
25	2020-02-12T03:34:23.508Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		64	5000	1
26	2020-02-12T03:34:25.953Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		17	1494	1
27	2020-02-12T03:35:21.573Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		112	8941	1
28	2020-02-12T03:36:08.382Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		29	2392	1
29	2020-02-12T03:36:09.153Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		15	1363	1
30	2020-02-12T03:36:13.930Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		27	2201	1
31	2020-02-12T03:36:14.992Z	AUTO				sand_stroke	["action":"sand_stroke","layer":"__AUTO__","hint_layer":"R1"]		22	1902	1
32	2020-02-12T03:37:59.101Z	ALL		0	0	draw_all	["layer":"__ALL__","angle_i:gesture=double_fist"]			95	1



A browser-based p5.js interface captures hand landmarks through a webcam and translates their position, orientation and movement into operational gestures. These gestures are converted into structured commands and transmitted through the control pipeline for robotic execution.

The interface turns bodily input into a direct control layer between the participant and the physical fabrication system.

Gesture -> Structured Command -> Robot Action

AI-DRIVEN RESPONSIVE MURAL

FINAL EXPERIENCE + PHYSICAL PROOF



Knowledge as Responsive Media

During daily teaching and chanting, spoken material and contextual input are processed through a live AI system and translated into evolving mural imagery. Rather than replaying a fixed recording, the visual surface changes in response to the knowledge being transmitted.



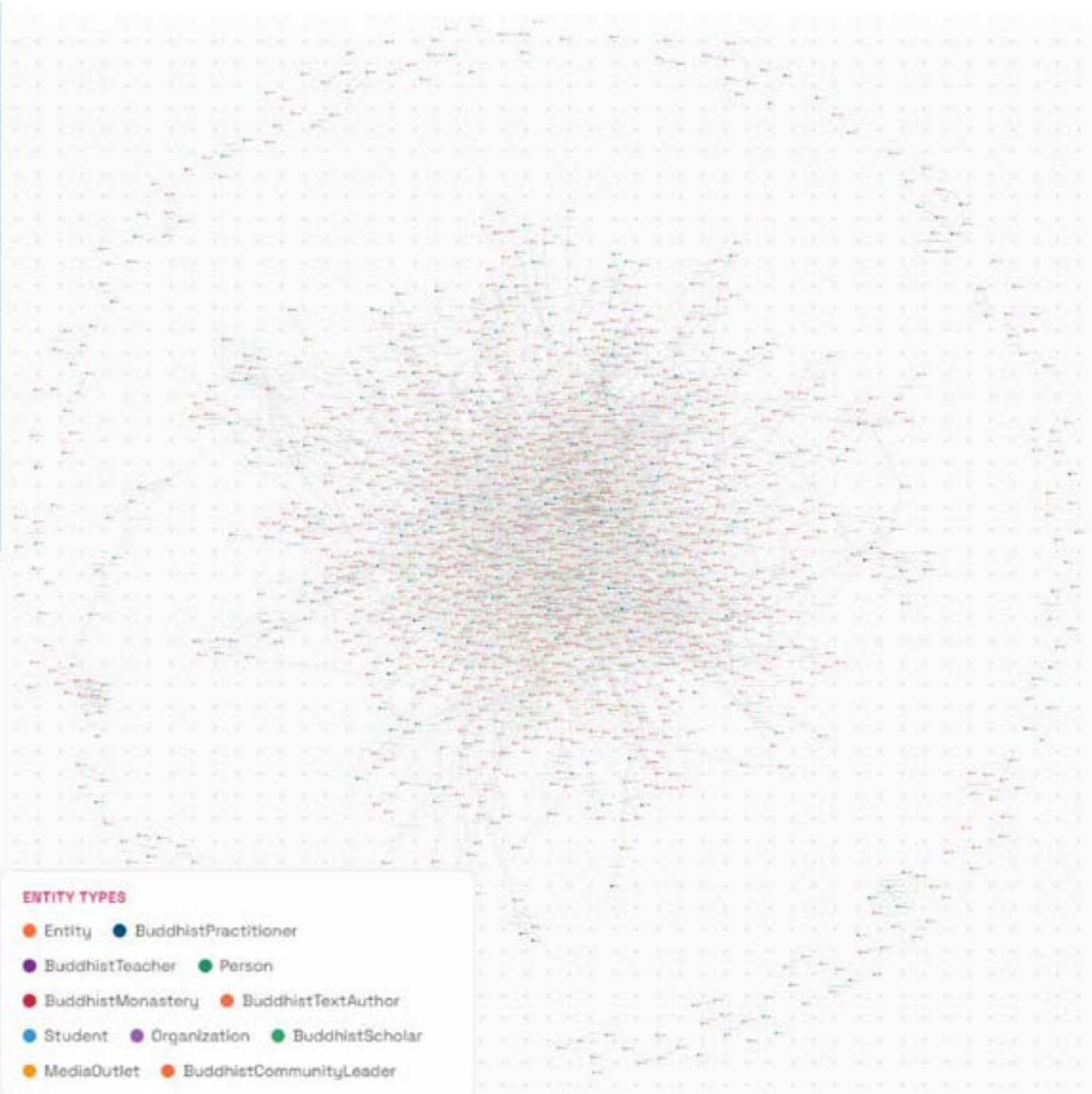
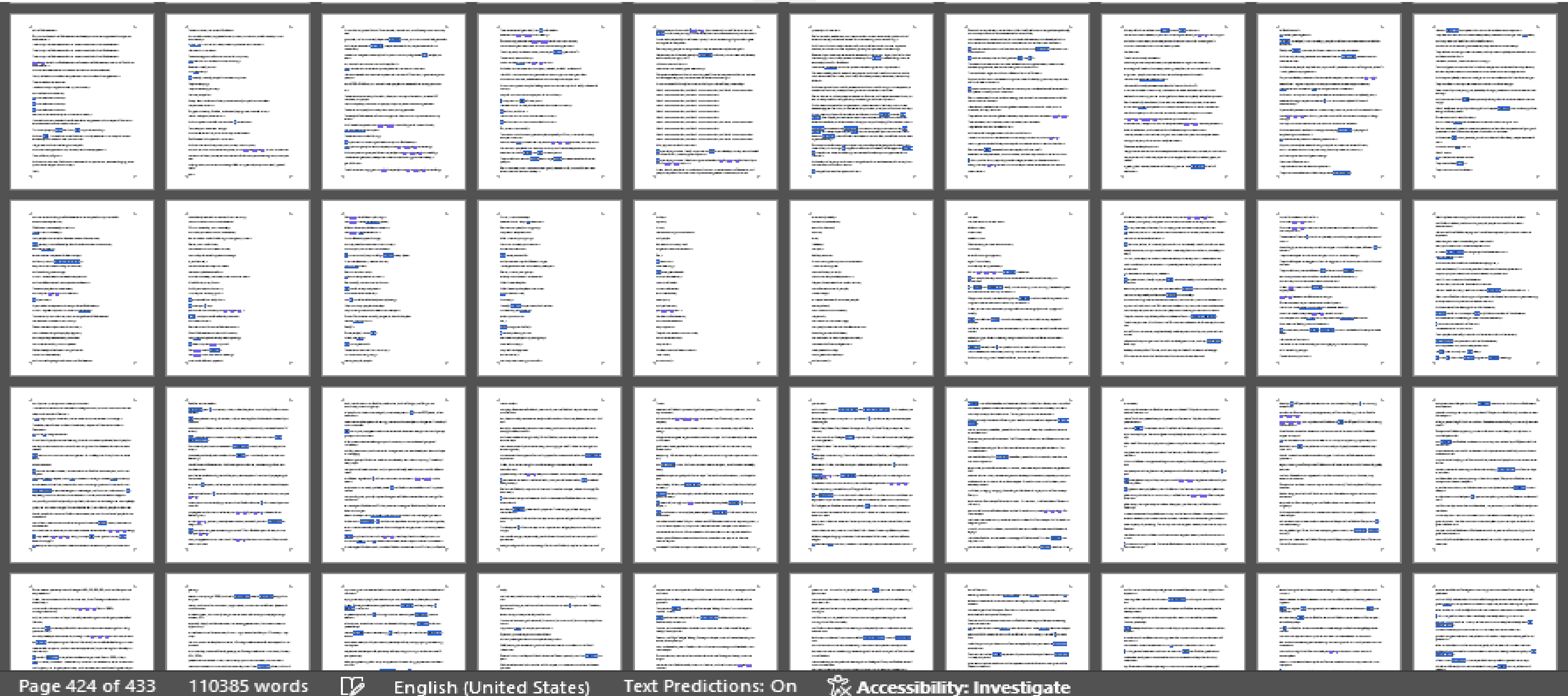
1:1 Responsive Media Prototype

The 1:1 prototype tests dynamically generated visual content as a physical media surface rather than a standalone screen. It demonstrates how responsive imagery can operate at full scale and become part of the surrounding environment.

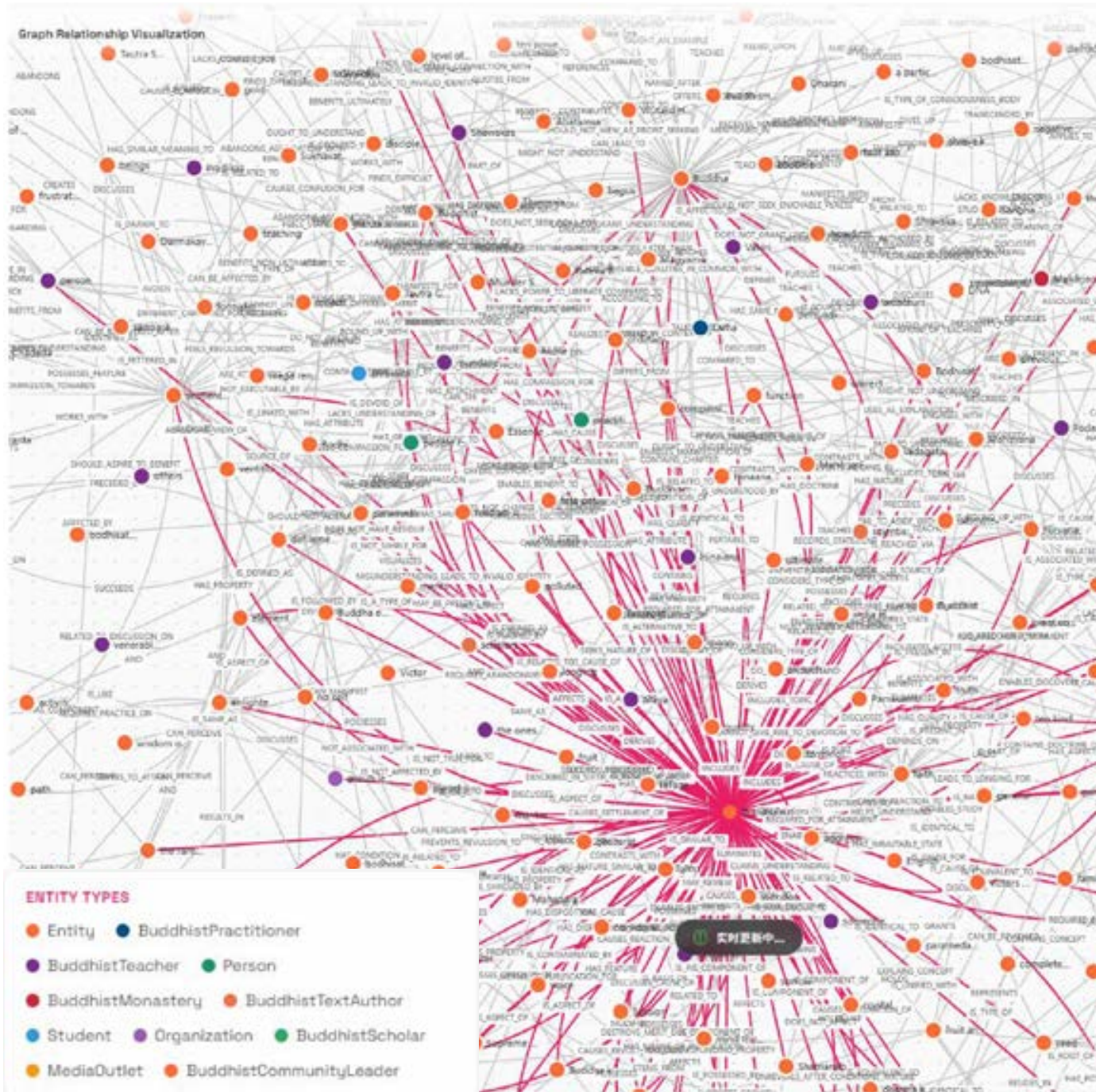
Building the AI Knowledge Archive

A corpus of public lectures by Khenpo Sodargye was collected, transcribed and translated into a machine-readable archive of more than 430 pages and 110,000 words, forming the primary knowledge source for the responsive media system.

430+ pages
110,000+ words



Source Material -> Transcription -> Structuring -> Retrieval -> Context



Zep-Based Knowledge Analysis

Document Ingestion
Concept Extraction
Entity & Relationship Analysis
Knowledge Structuring

The structured archive is analysed to identify recurring concepts, entities and relationships, providing retrievable context for the downstream AI system.

From Knowledge to Live Media

The final system operates as a modular Python application coordinating input capture, contextual retrieval, LLM reasoning, visual-state selection, Seedance generation, post-processing and live playback within a unified runtime.

Input Capture
Context Retrieval
Reasoning
Visual Selection
Generation
Post-Processing
Live Playback

Custom Python-based AI agent
and media pipeline

```
reasoner.py X
reasoner.py > ...
60 def format_evidence_block(ctx: dict) -> str:
61     src = e.get("source_name", "") or e.get("source", "") or "?"
62     rel = e.get("name", "") or "RELATES_TO"
63     tgt = e.get("target_name", "") or e.get("target", "") or "?"
64     fact = (e.get("fact") or "").strip().replace("\n", " ")
65     line = f"- {src} --{rel}--> {tgt}"
66     if fact:
67         line += f" | fact: {fact[:180]}"
68     parts.append(line)
69
70 if matched_episodes:
71     parts.append("\nEvidence passages:")
72     for i, ep in enumerate(matched_episodes, 1):
73         text = (
74             ep.get("text")
75             or ep.get("content")
76             or ep.get("summary")
77         )
78
79 PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\98371\Desktop\AI_Buddhism> python run_buddhist_reasoner.py
=== ANSWER ===
One-sentence essence:
The pursuit of wealth is a human preoccupation that can be understood through the lens of
Direct answer:
The path to wealth is often fraught with the desire for more, which can lead to attachmen
asting values.
Buddhist framing:
In Buddhism, the desire for wealth can be seen as a manifestation of craving, one of the
Cultivating compassion and understanding the interdependent nature of all phenomena can h
```

- display_player
 - __pycache__
 - __init__.py
 - play_history_then_live.py
 - watch_and_postprocess.py
 - watch_display_loop.pyManages live playback and updates
- input_gateway
 - __pycache__
 - audio
 - clients
 - input_logs
 - server.py
 - settings.py
 - storage.py
 - transcribe_whisper.pyCaptures and transcribes input
- pipeline
 - __pycache__
 - outputs
 - run_live_pipeline.py
 - test_latest_pipeline.pycoordinates system logic and data flow
- reasoner
 - __pycache__
 - image_selector.py
 - packet_reader.py
 - test_directed_selection.py
 - test_latest_packet.py
 - test_select_image.py
 - visual_state_reasoner.pyretrieves context and selects visual states
- runtime_logs
 - display_loop.log
 - input_gateway.log
 - live_pipeline.log
 - s3_ambient.log
 - seedance_watcher.log
 - watch_and_postprocess.logMonitors system execution
- video_generation
 - __pycache__
 - keyframes
 - seedance_outputs
 - state
 - .env
 - 000_start.png
 - postprocess_latest_video.py
 - seedance_from_pipeline.py
 - seedance_interactive.py
 - watch_seedance_from_pipeline.pySends prompts to Seedance and post-processes outputs
- visual_library
 - images
 - mural_index.json
 - .txtOrganises indexed visual references and mural states
- system_launcher_final.py

Visual Library

Mural imagery documented at Gongga Qude Monastery was organised into indexed visual states, providing the reference library for the generative media system.

While the Robotic Sand Mandala translates structured geometry into physical fabrication, the Responsive Media system transforms archived knowledge into evolving visual content.

Arising



Abiding



Dissolution



Emptiness



Transformation



Practitioner Feedback

Prototype Review & Testing in Context

Buddhist practitioners reviewed and tested the prototypes during fieldwork, providing direct feedback on their operation, cultural appropriateness and relationship to existing ritual practice. Their feedback helped refine how the technology should support embodied practice rather than replace it.



BANDIAN DUNYU - VICE PRESIDENT · TIBETAN BUDDHISM ASSOCIATION OF CHINA



VENERABLE WEISHAN - BUDDHIST SCHOLAR · RENMIN UNIVERSITY OF CHINA, BEIJING

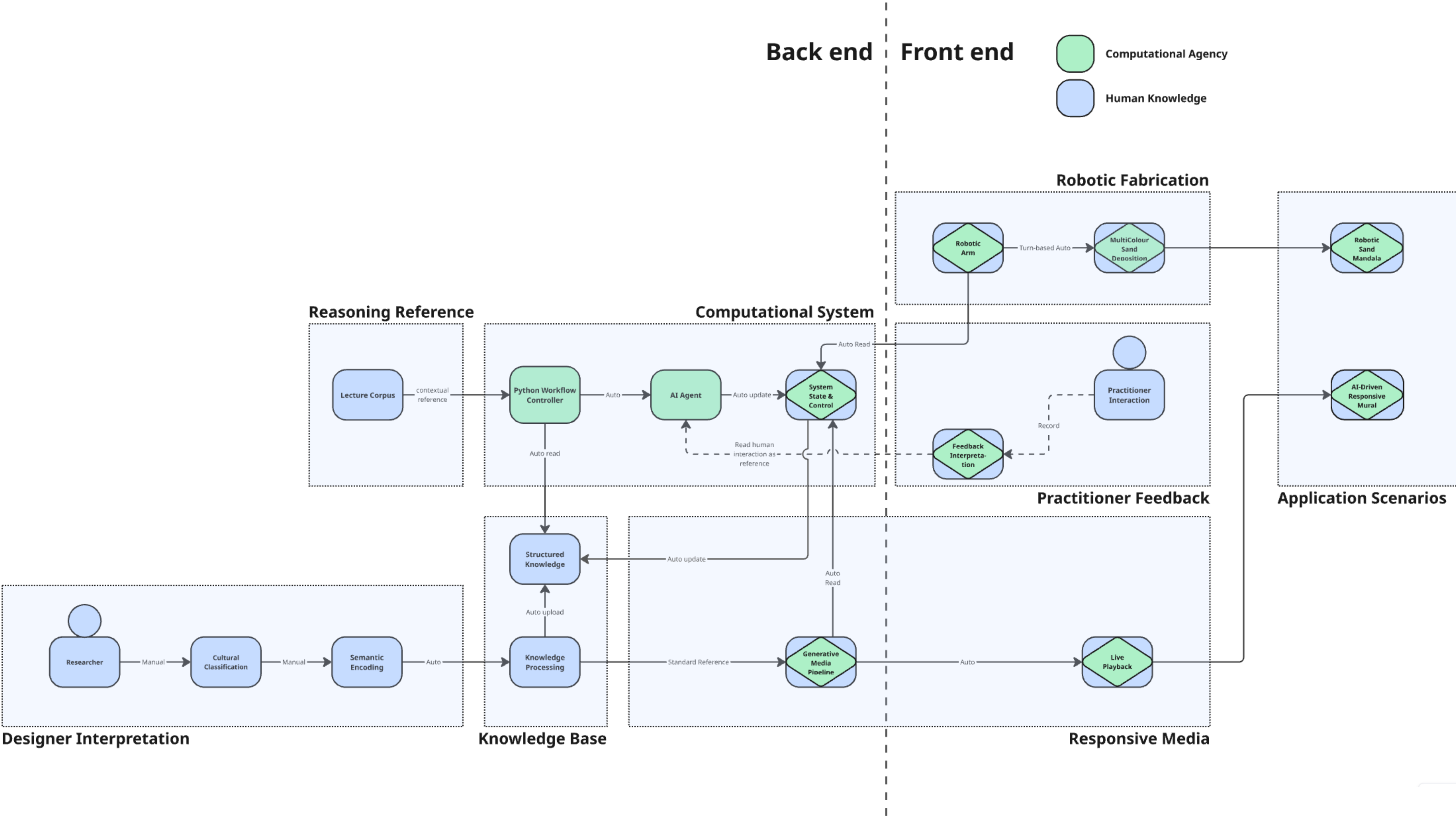


GESANG QUNPEI RINPOCHE - ABBOT, GONGGA QUDE MONASTERY, TIBET

System Overview

From Cultural Knowledge to Two 1:1 Applications

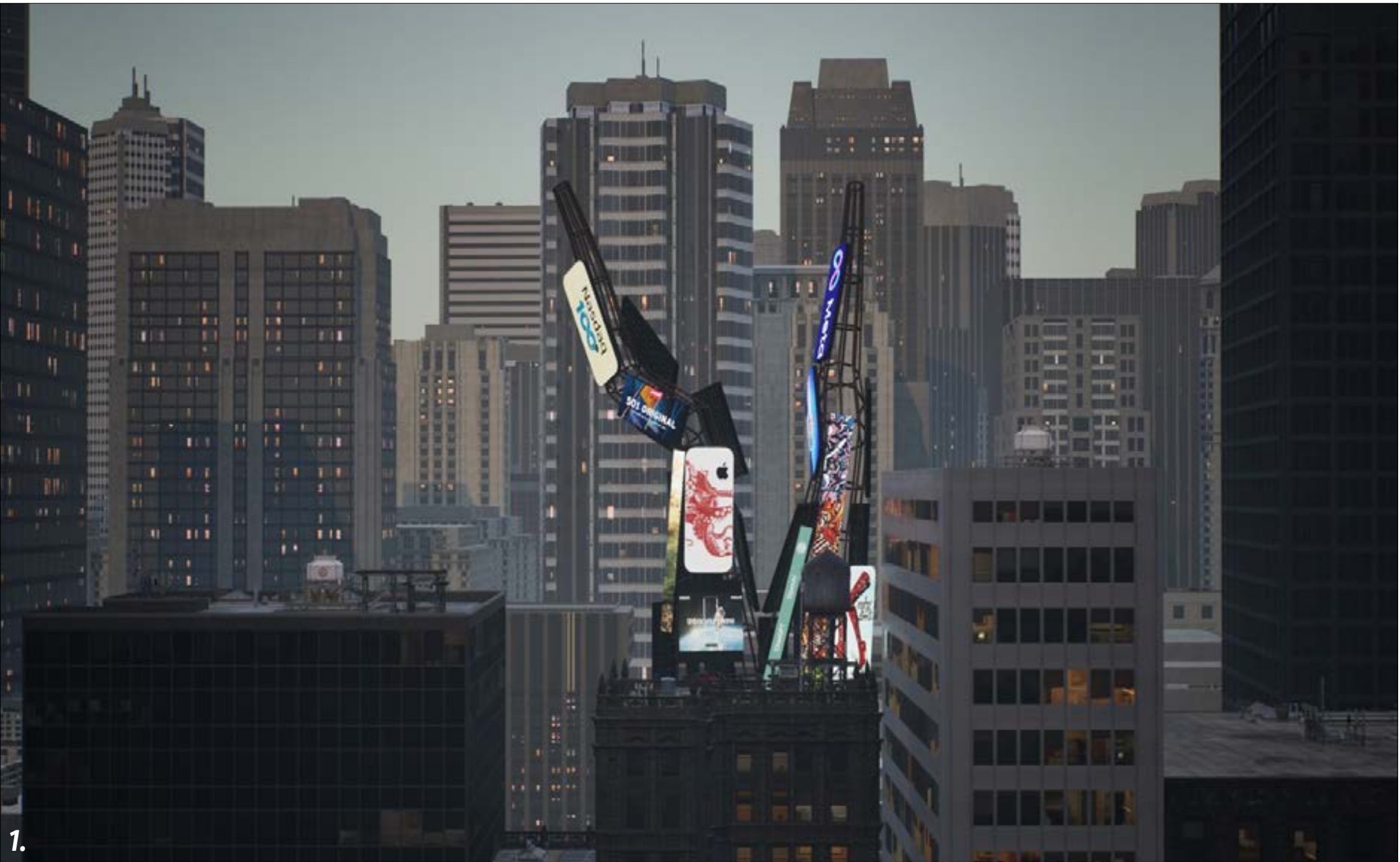
The two prototypes operate as parallel applications within a shared physical–digital system. Designer interpretation, structured cultural knowledge and computational processing are translated into either robotic fabrication or responsive media, while practitioner feedback is returned to the system to inform further refinement.



2. LIGHT UP NEW YORK FOR BUT A MOMENT

A TRANSFORMABLE MEDIA STRUCTURE
FOR NEW YEAR'S EVE

The project reimagines Times Square's dispersed advertising surfaces as a single transformable media structure. Moving through the city over the course of New Year's Eve, the structure shifts between four spatial, commercial and performative identities before becoming the ball itself at midnight.



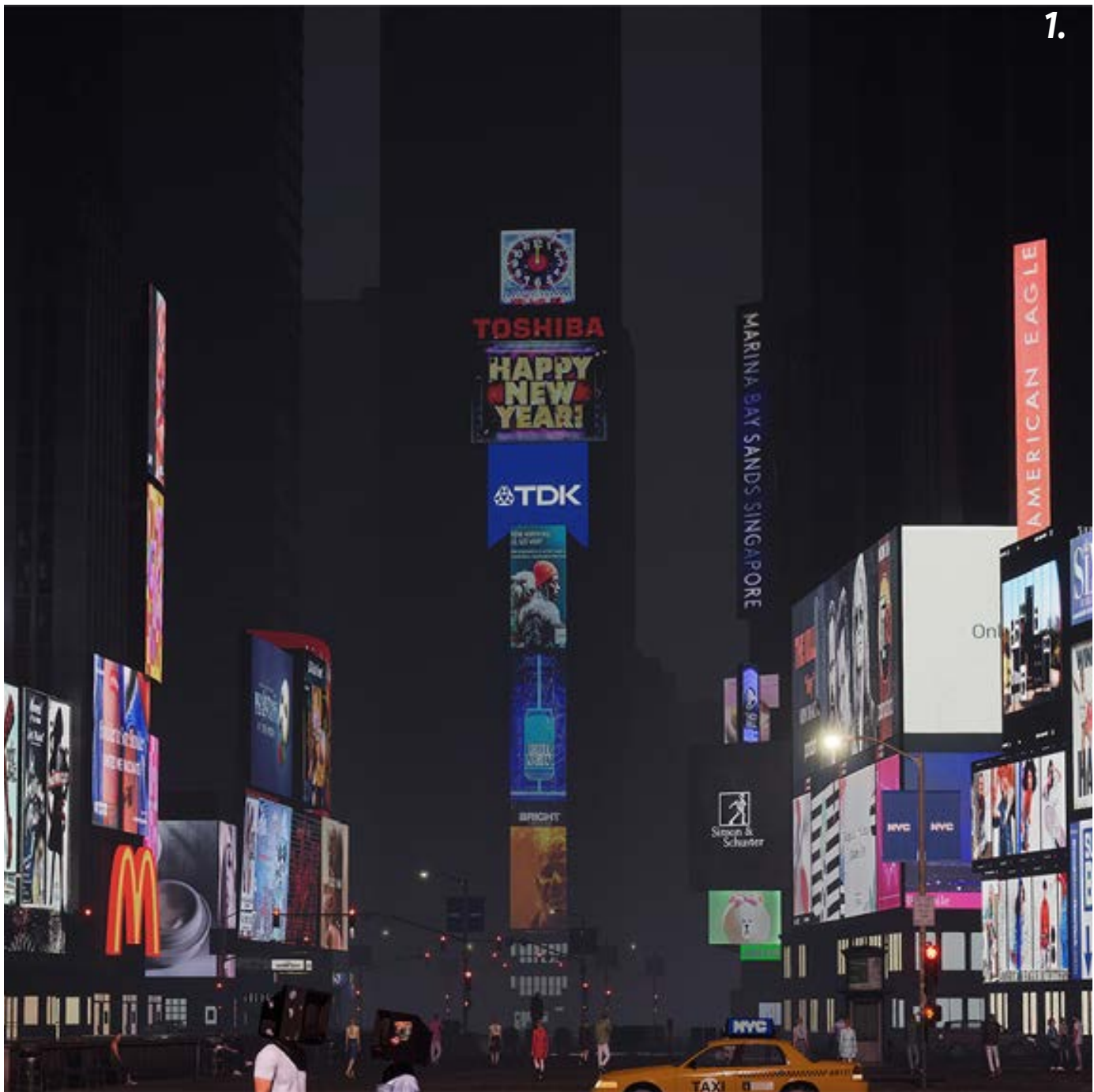
- 1. Chthulucene**
An unfamiliar moving media spectacle concentrates the city's attention into a single object.
- 2. Big-Mac**
The structure reforms into a monumental commercial identity, temporarily turning branding into inhabitable urban form.
- 3. Coexistence**
The media body opens into a public stage, shifting from commercial display to collective performance.
- 4. Light Up New York for but a Moment**
At midnight, the previous identities disappear as the structure becomes the New Year's Eve ball and completes its journey to One Times Square.



youtu.be/lgz9JANOXDg Final film - 8 mins

Media Breaks Free — From Façade to Moving Structure

As digital media increasingly conceals the architecture of Times Square, the project proposes a reversal: billboard surfaces are detached from the façades and concentrated into a single mobile structure. Once released, media is no longer a fixed surface but a moving body with its own structure, depth and spatial presence.



1. Façade Saturation

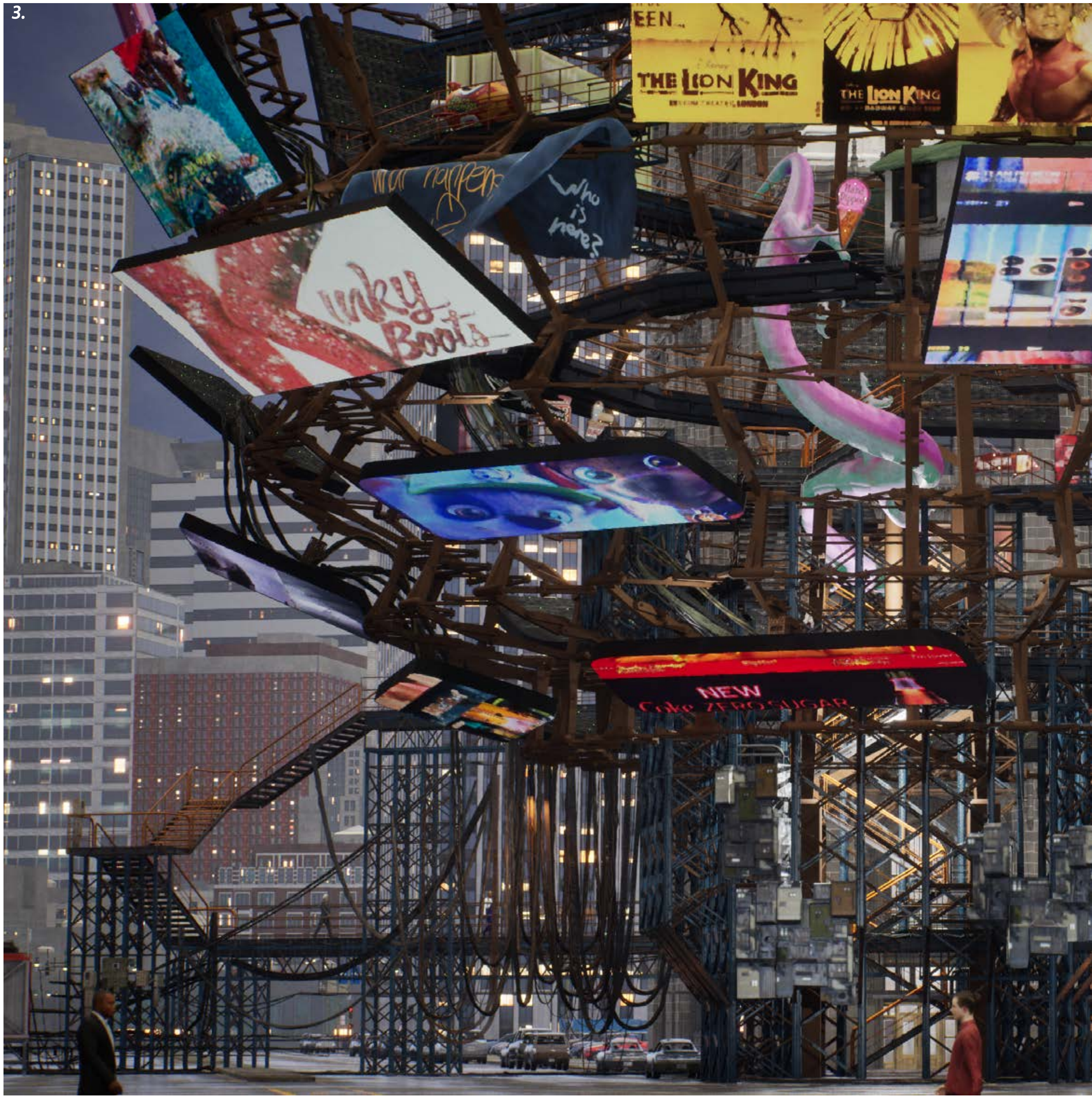
Media increasingly dominates the surrounding buildings.

2. Media Concentration

Dispersed billboard surfaces are detached and assembled into a single mobile body.

3. Media Takes Form

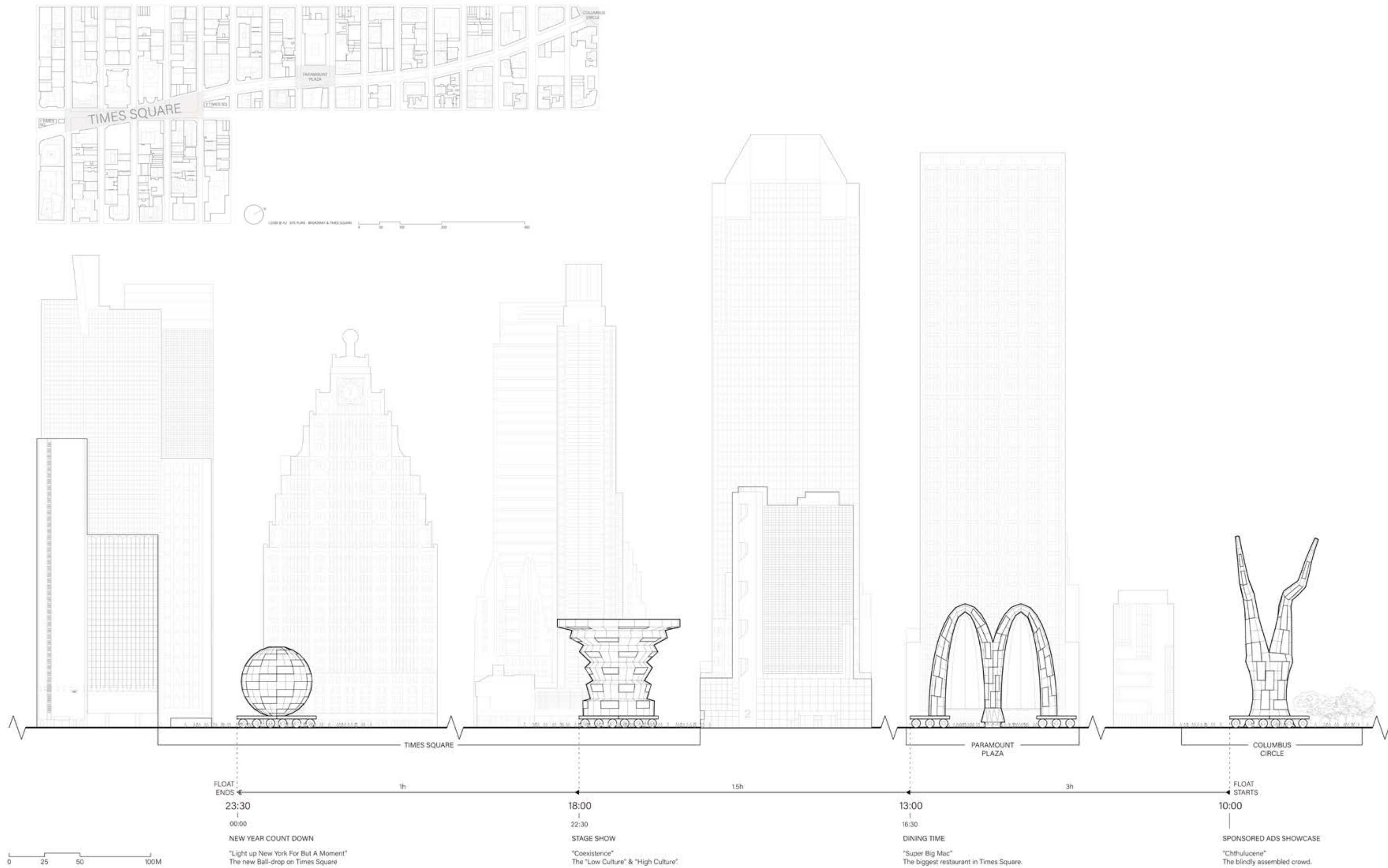
The billboard acquires structure, depth and interior space, becoming part of the parade itself.



Media Takes Form
— A Transforming Parade Along Broadway

The New Year's Eve Ball Drop is reimagined as a moving parade rather than a single event at midnight. Throughout the day, the media structure travels along Broadway, transforming between commercial, performative and public identities before its final arrival at One Times Square.

Each transformation changes not only its appearance, but also how the structure is occupied, viewed and used along the route.



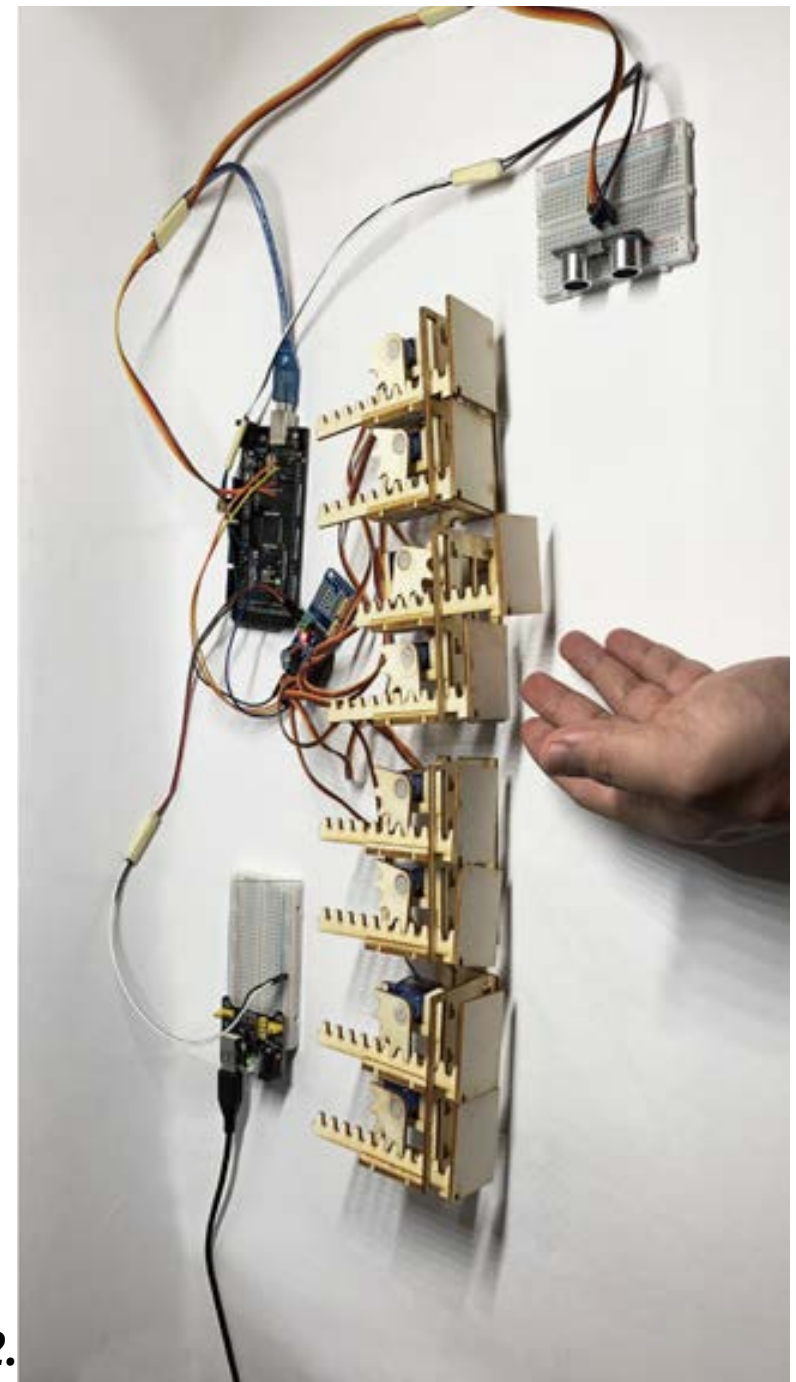
Structure, Movement, Form
— Kinetic Prototyping & Structural Iteration

A series of physical studies tested how a single structural system could shift between different configurations. Form models explored alternative transformation geometries, while an Arduino-controlled prototype tested movement and mechanical response at physical scale.

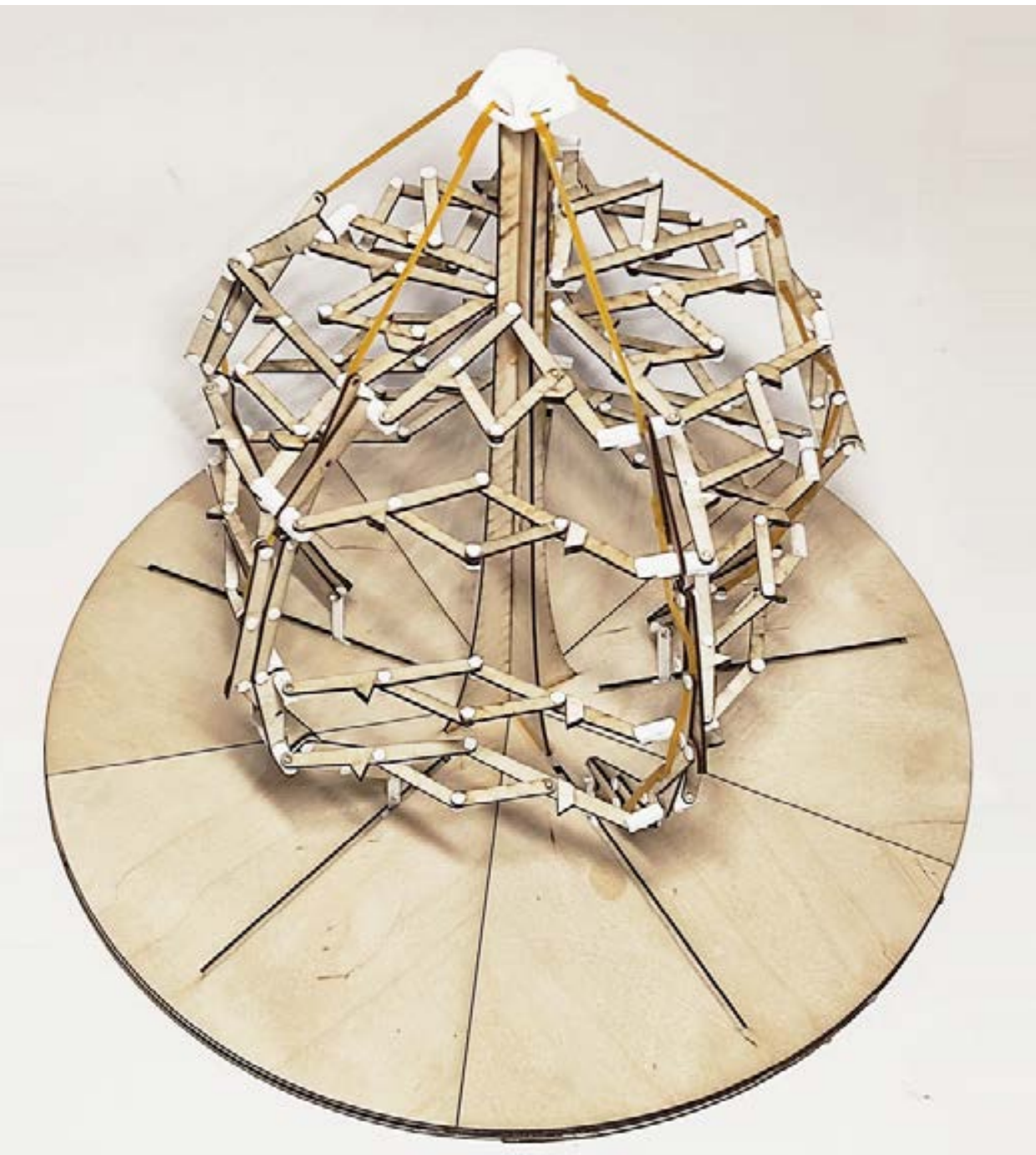
These experiments were translated into an extended structural study showing how the same primary frame reconfigures across the different stages of the parade.



1.



2.



1. Form Iteration

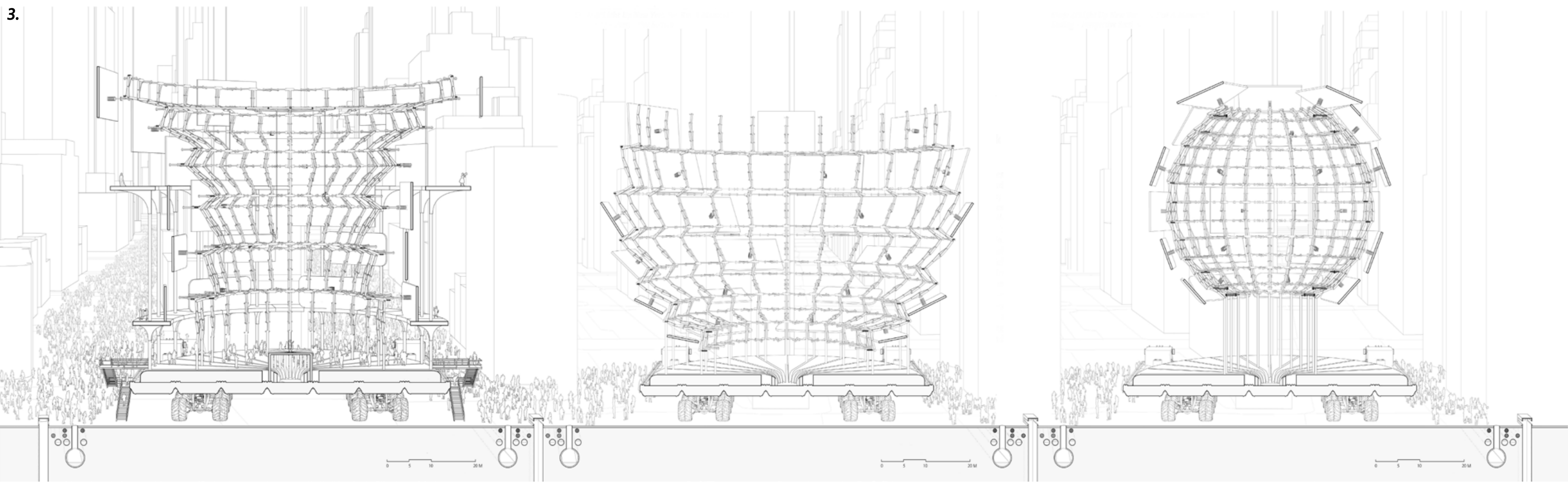
Alternative geometries and transformation states.

2. Kinetic Prototype

Physical testing of movement and mechanical response through an Arduino-controlled transformation prototype.

3. Structural Translation

Scaling the tested transformation logic into a full structural system.



3.

3. NO CURTAIN CALL

IMMERSIVE THEATRE BETWEEN
PHYSICAL AND VIRTUAL SPACE

No Curtain Call was developed as an interactive theatre experience existing simultaneously across physical and virtual space. A shared game map connects the real performance environment with its digital counterpart, allowing audiences to move between live theatre, tactile spatial interaction and virtual scenes.

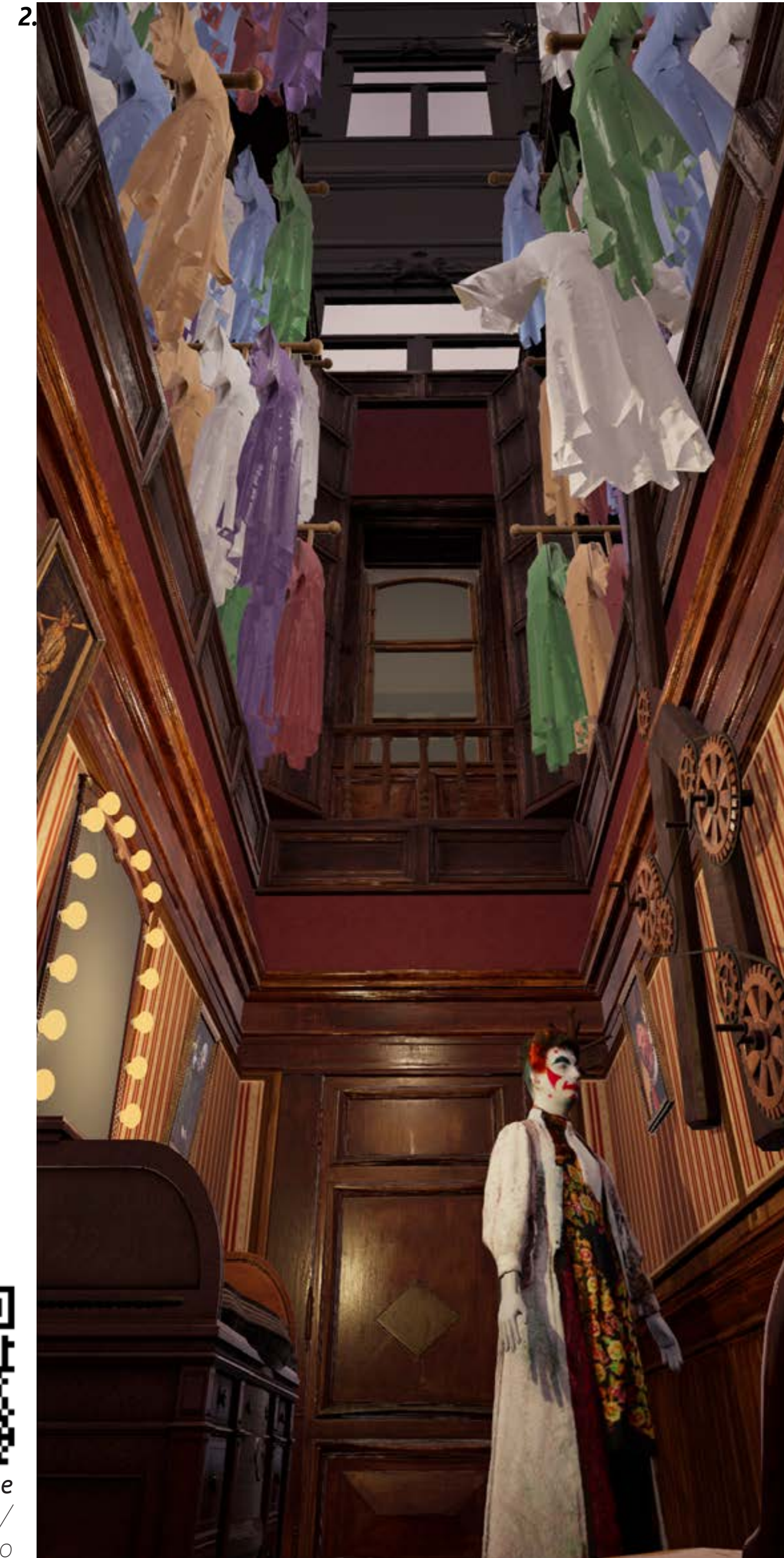
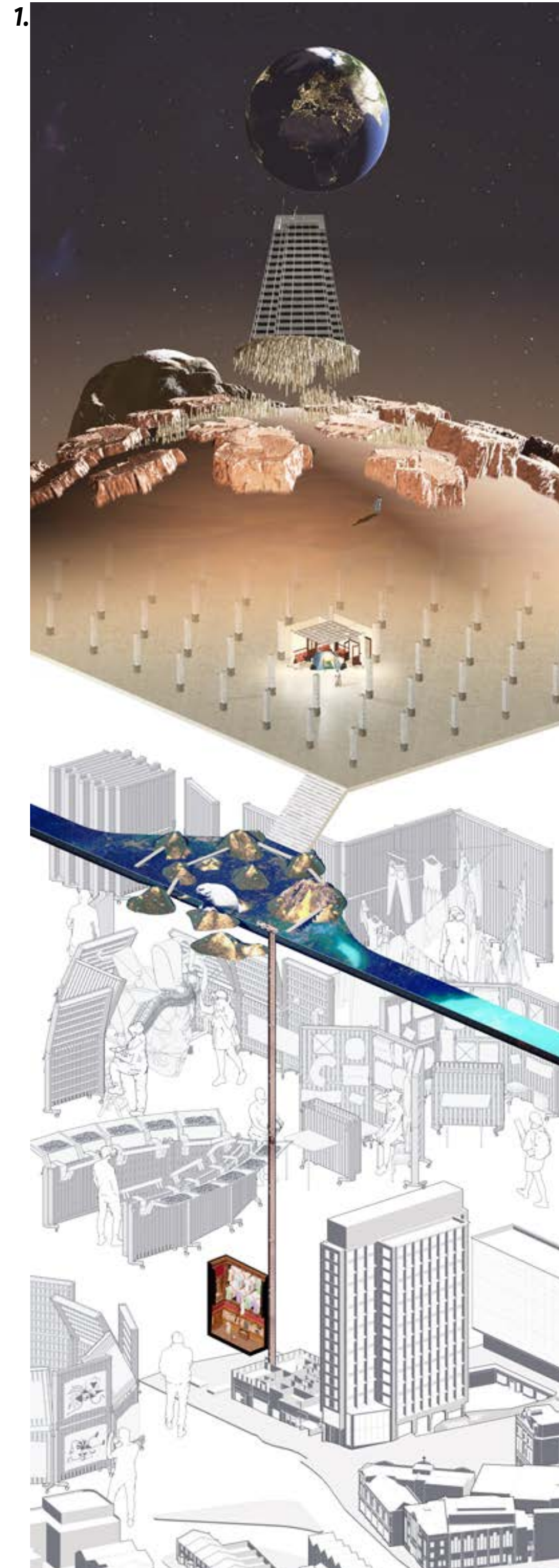
The physical environment was built from modular reclaimed bed-frame elements, creating movable boundaries and stage components that could be reconfigured throughout the performance.

1. Physical–Virtual Game Map

The game map defines a theatre operating simultaneously across physical and virtual space, linking audience movement, live performance and digital environments within a single spatial journey. Modular elements made from reclaimed bed frames form the interactive physical counterpart to the virtual world.

2. Selected Performance Scenes

A series of scenes developed across the shared physical–virtual environment, combining live performance, spatial interaction and digital media.



VR Performance
youtu.be/
FP47Lg52VHo



Building the Performance System

Physical performance, digital reconstruction, motion capture and object tracking were combined into a single real-time performance workflow.

1.



1. Physical Performance Preparation

Costume, make-up and stage identity developed in physical production.

2. Character Digitisation — Digital Twin

The performer was reconstructed as a digital counterpart.

3. Inertial Motion Capture

Live movement transferred into Unreal Engine in real time.

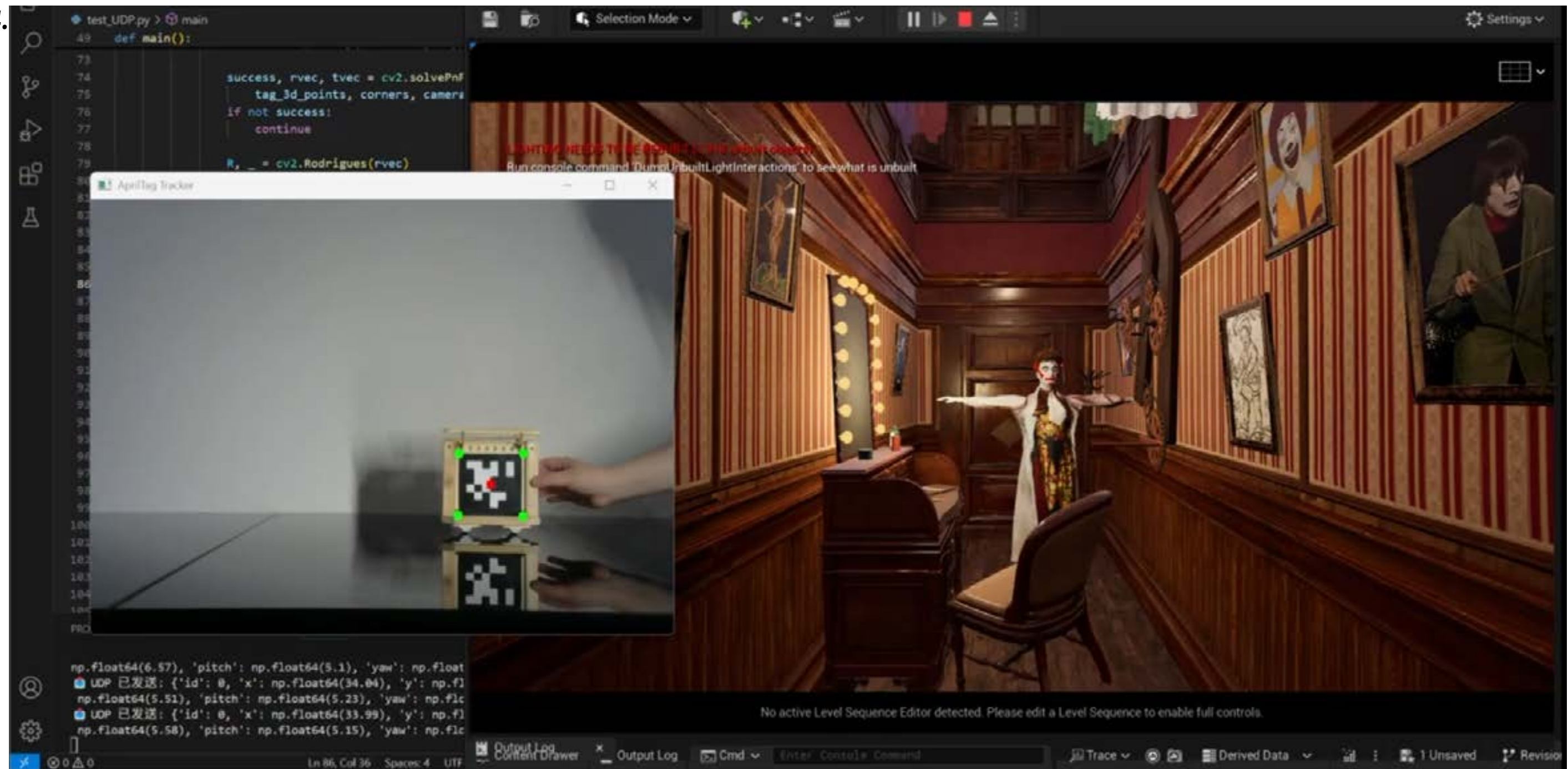
4. AprilTag Object Tracking

Physical stage elements mapped directly into the virtual environment.

2.



4.



3.



Testing Video
youtu.be/
BJPKWB3YkVE

4. MAKING ACROSS SCALES

SCULPTURE
 MATERIAL EXPERIMENTATION
 MODEL MAKING
 PHYSICAL PROTOTYPING



Hands-on making has remained central to my design process, from figurative sculpture and material experimentation to structural models and physical prototypes.

5. PROFESSIONAL PRACTICE

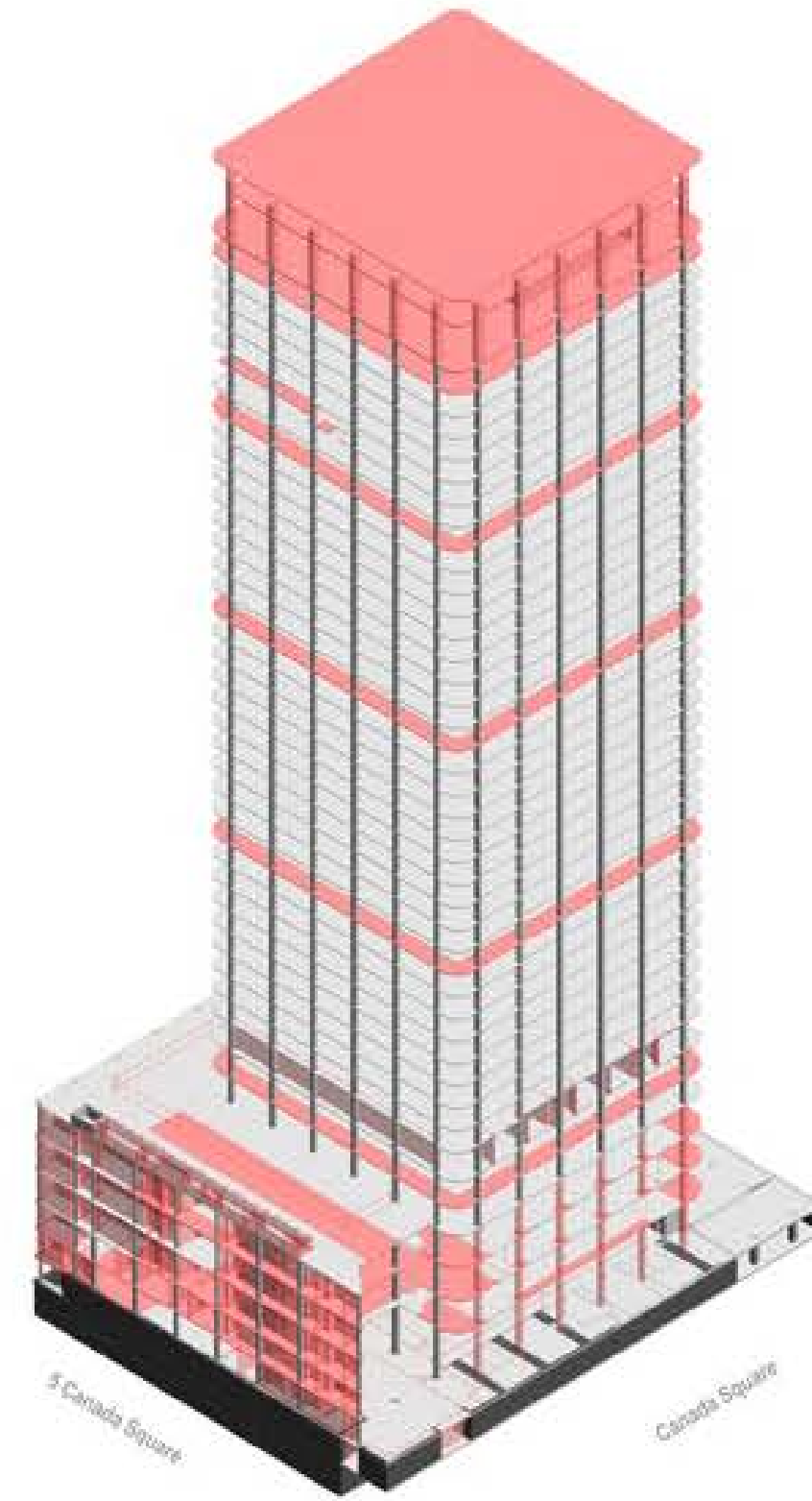
FOSTER + PARTNERS · STUDIO 1 · LONDON

8 CANADA SQUARE

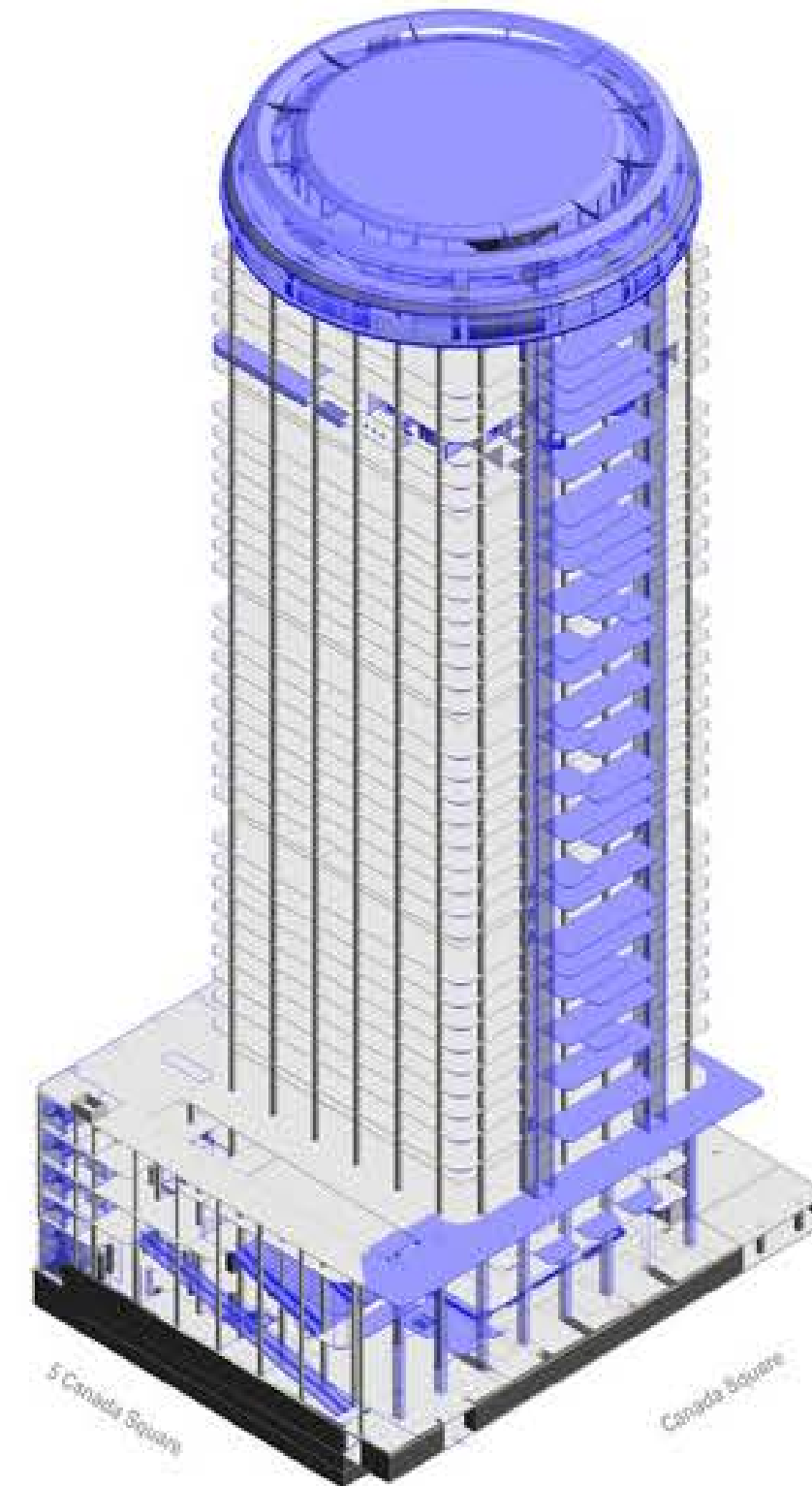
MODEL COORDINATION & REAL-TIME VISUALISATION

Developed as a competition proposal, 8 Canada Square explored the transformation of the existing tower through selective retention, demolition and new intervention. I took primary responsibility for the Revit model, coordinating these changing conditions within a single structured digital environment.

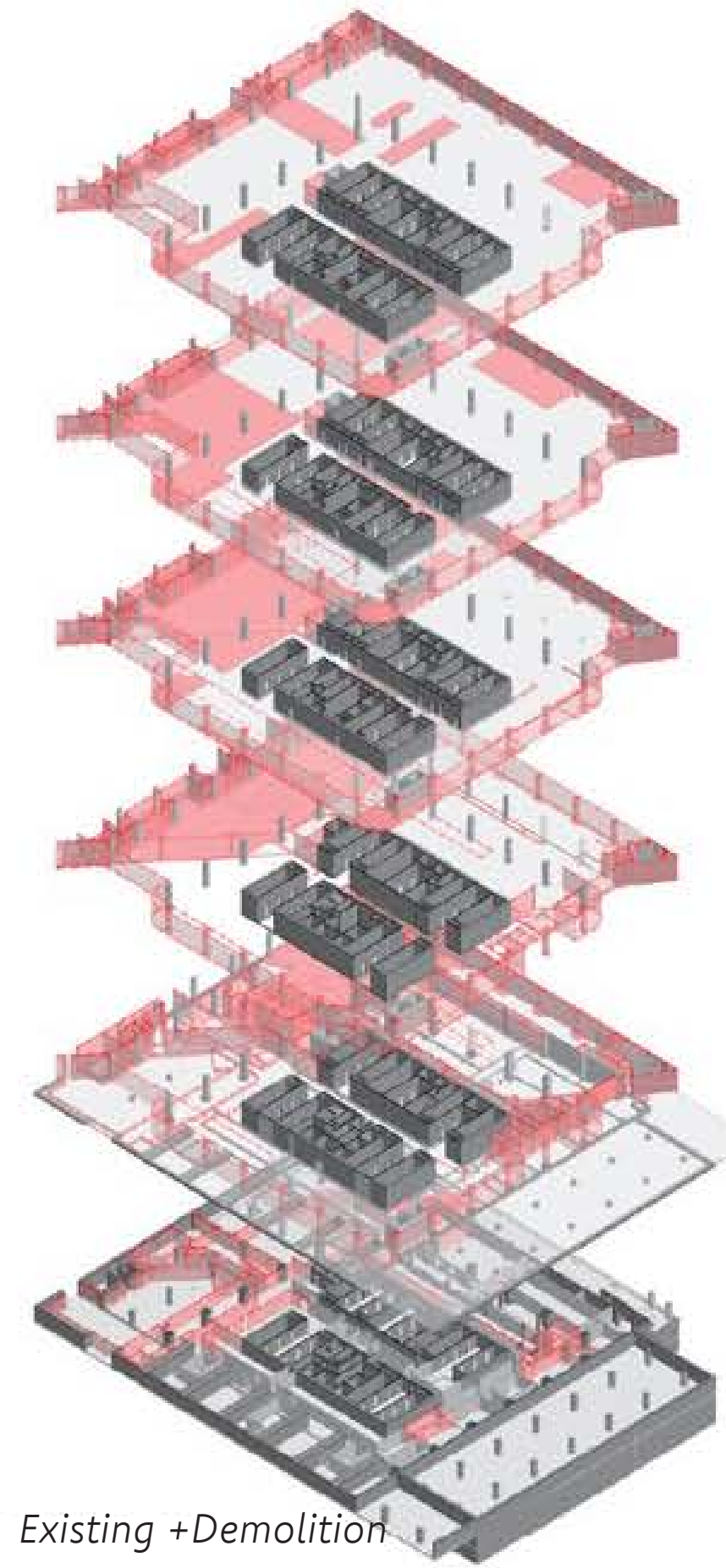
Alongside the architectural model, I developed the cinematic visualisation workflow for the Tower Top exhibition space, translating coordinated design information into real-time environments in Unreal Engine.



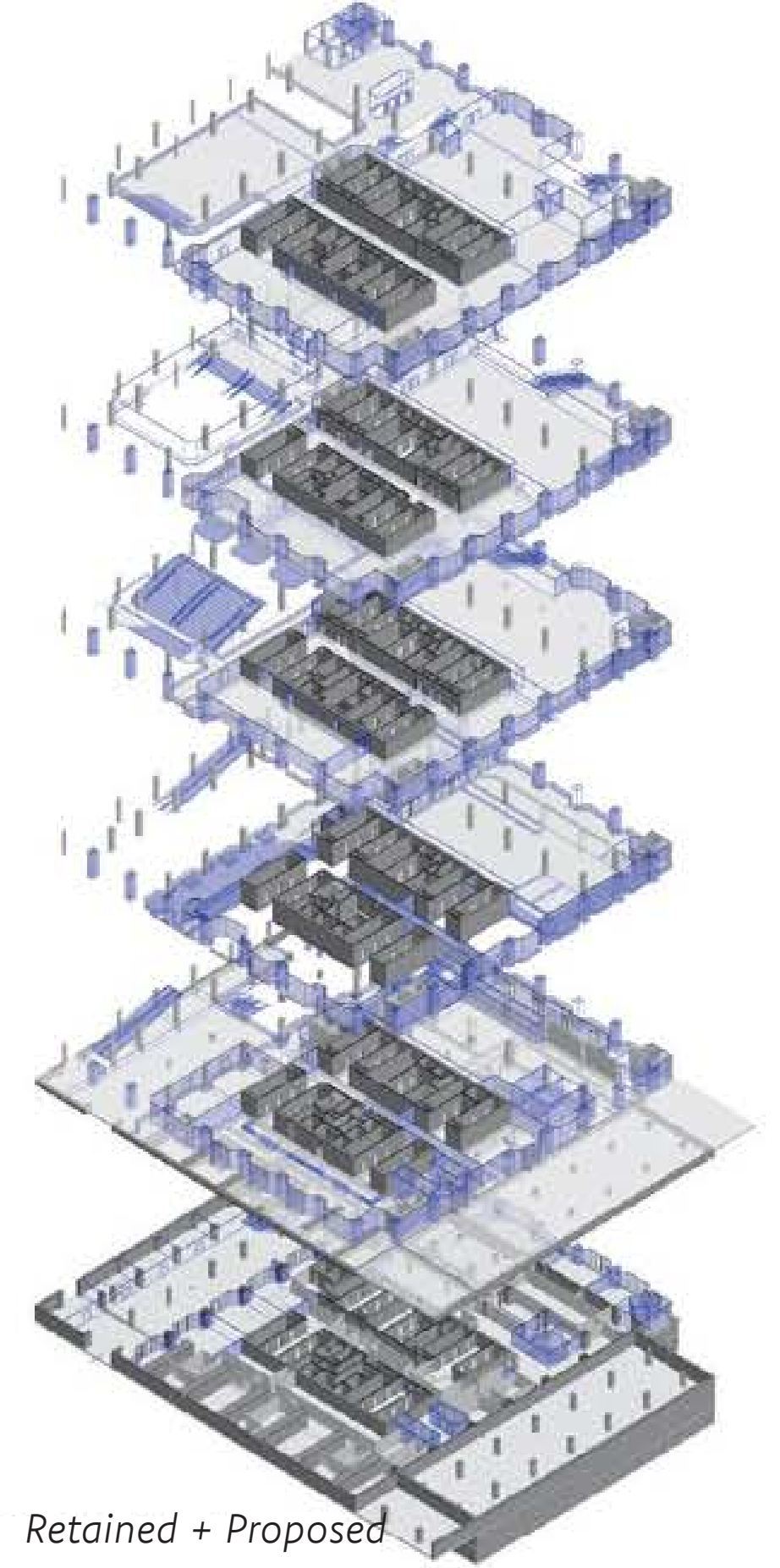
Existing + Demolition



Retained + Proposed



Existing + Demolition



Retained + Proposed

Tower Top — Real-Time Visualisation



1 SPITALFIELDS

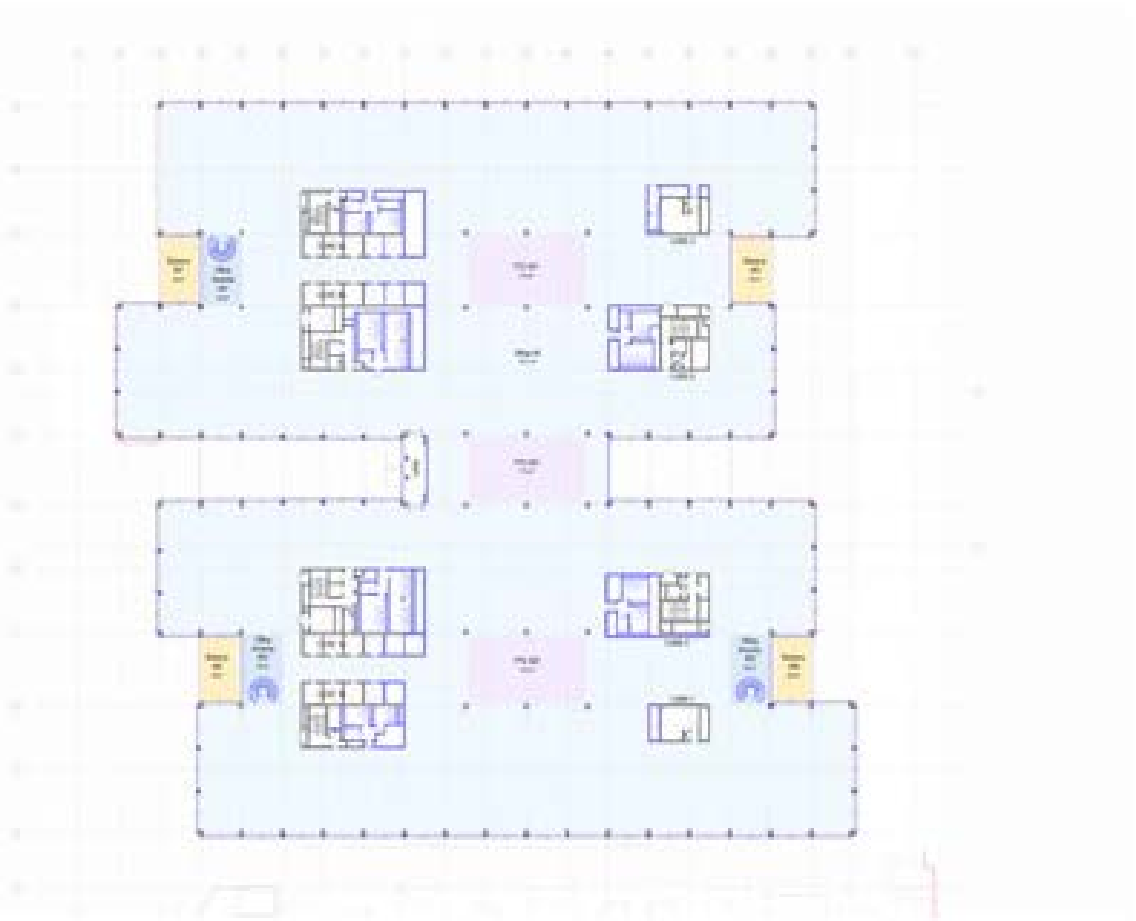
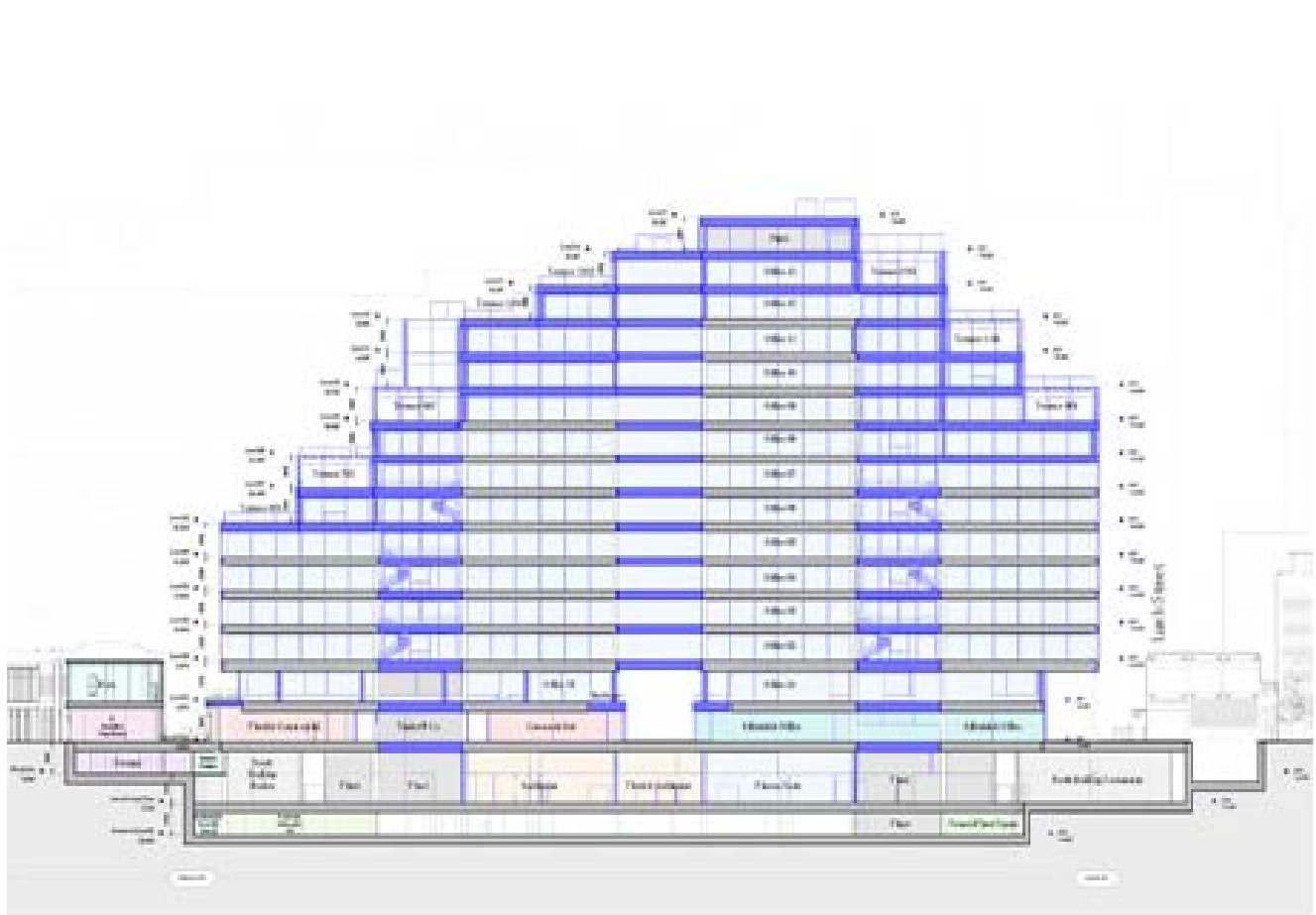
REVIT DELIVERY & TECHNICAL COORDINATION

I worked on 1 Spitalfields through its transition from RIBA Stage 2 to Stage 3, contributing to the development of the main building, market and affordable workspace within a complex historic urban context.

A significant part of my role involved translating evolving design decisions into coordinated Revit information. I produced and coordinated general arrangement and detailed drawing packages, including plans and sections across existing, demolition and proposed conditions, while working with the wider project team to maintain consistency between design development and technical delivery.



Main Building - Proposed Perspective



Main Building — Revit Drawings



Market - Proposed View I



Market - Proposed View II